

OpenAI's math release: highlights and insights

What is in the **openai/math** repository (722 manuscripts produced by an unreleased internal OpenAI model), which claims are machine-checked, and how to read them, explained from first principles.

Manuscripts 722 grouped into result families	Result families 372 across 17 subjects	Problems posed to the model ~4,000 ≈3 h of thinking per result
Families with a Lean proof 235 63% of families; 330 of 722 papers (46%)	Machine-checkable statements 405 0 "sorry"; only standard axioms	Lines of Lean code 25.9M in 121,735 files (Mathlib ≈ 2M)

The short version

- **It is genuine.** The repository belongs to the OpenAI GitHub organisation, and your fork (oishik1306/math) matches it exactly.
- **The claims are extraordinary.** They include a zero-free half-plane for the Riemann zeta function, the Unique Games Conjecture, a matrix-multiplication exponent $\omega \leq 9/4$, the Mahler conjecture, the Birch–Swinnerton-Dyer formula in ranks 0 and 1, the Hodge conjecture for CM abelian varieties and the isomorphism of free group factors.
- **Much more is formally checked than first appeared.** 235 of 372 families (63%) come with Lean proofs, including the zeta zero-free half-plane, Unique Games, π 's irrationality exponent and the free group factors.
- **But Lean often checks less than the title says.** In 59 formalised families the headline claim itself is not what Lean proves (Section 6 lists every one). The famous arithmetic-geometry claims (BSD, Hodge, modularity) have no formal proof at all.
- **The bottleneck is now human understanding.** Hundreds of manuscripts appeared within days, far faster than mathematicians can referee them.

About this revision. The first edition (6 October) reported Lean proofs for only 127 families. It relied on `lean/formalization.yaml`, which lists just 178 of the 405 Lean challenges. This edition uses OpenAI's per-family pages in `lean/docs`, cross-checked against an independent statement-by-statement review (Section 8). It also adds a physics section and a gap audit.

How to use this guide: the contents page and the PDF bookmarks are clickable. In Section 4, each row of the summary table jumps to an explainer card. Appendix A lists all 372 families with links to their papers. Badges:

 LEAN-CHECKED	Lean proves the headline claim	 PARTLY LEAN	Lean present, but proves less than the headline
 PAPER ONLY	no formal proof; an unrefereed paper		

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1 Is it really from OpenAI?

Yes. I checked the upstream repository directly and compared it with your fork:

Check	Result
Owner of github.com/openai/math	The OpenAI GitHub organisation
History and licence	A single commit, Apache-2.0 licence
Your fork, oishik1306/math	Identical README text and top-level layout (lean/, preprints/, reasoning_traces/, overview.pdf, CONTENTS.md), and the same single “Initial commit”
Internal metadata	overview.tex is dated 6 October 2026 with PDF author “OpenAI”; the Lean catalogue lists “OpenAI” as author
Press coverage	Several outlets report the release of 722 manuscripts in 372 families on 6 October 2026

Where this release fits

This is OpenAI’s third mathematics release in about ten weeks. The context helps when judging the claims.

Date	Event
Oct 2025	An OpenAI executive says GPT-5 “solved” 10 Erdős problems. The maintainer of the Erdős-problems website calls this a misrepresentation: the model had found existing papers. The episode is a reason to read claims carefully.
1 Aug 2026	Ten results with Lean certificates, headlined by the first explicit non-sofic group.
Aug 2026	arXiv 2608.14673, “A Human Audit of OpenAI’s AI-Generated Mathematical Proofs”, reviews those ten results. It finds no confirmed substantive error in a principal result, but notes uneven review depth and flags one chapter for major revision. It does <i>not</i> cover this repository.
8 Sep 2026	A 166-page Navier–Stokes finite-time blowup proof with Lean files. Mathematicians broadly accepted it because the Lean check passed, but called the paper very hard to read, and there was a public dispute about credit. That proof is <i>not</i> in this repository; family 376 here is a different Navier–Stokes result.
21 Sep 2026	Terence Tao announces an Advisory Group on Mathematics and AI. Its remit includes advising on how to coordinate a large release of AI-produced results.
6 Oct 2026	This repository is released. The same day, Tao announces “Hexagon”, a repository for research works prompted by the flood of LLM-assisted results. The README says OpenAI is “exploring community-hosted repositories”; whether that means Hexagon is my guess, not stated.

Note: arXiv, Tao’s blog and several news sites were blocked from my working environment, so details of the items above come from search-engine summaries of those pages.

2 The collection in numbers

According to the README, the model was posed about **4,000 open problems** as part of OpenAI's model evaluations, which were expanded after existing maths benchmarks “saturated”. Each result used on average about three hours of ChatGPT Pro-level thinking. OpenAI grouped the outputs into families (a main result plus companions, consequences or alternative proofs) and kept those judged significant. That gives 372 families and 722 manuscripts. Since families are not one-to-one with problems, 372 out of 4,000 is not a clean success rate.

Families by subject, and how many have a Lean proof

Subject	Families	with Lean		Share	Papers	with Lean
Theoretical computer science	40	32		80%	73	54
Combinatorics	37	33		89%	50	39
Algebraic and complex geometry	36	7		19%	89	9
Number theory	31	16		52%	58	17
Probability and statistical mechanics	29	19		66%	105	40
Differential geometry	29	15		52%	49	18
Mathematical physics	25	17		68%	59	23
Operator algebras	19	14		74%	31	15
Algebra	18	9		50%	29	13
Topology	18	3		17%	26	3
Real and complex analysis	16	9		56%	26	11
Partial differential equations	16	11		69%	29	14
Convex and metric geometry	15	13		87%	30	24
Group theory	14	12		86%	22	16
Dynamical systems and ergodic theory	12	9		75%	19	9
Functional analysis	11	10		91%	19	18
Mathematical logic	6	6		100%	8	7
All subjects	372	235		63%	722	330

Source: OpenAI's per-family Lean pages (lean/docs/NNN.md), which name each family's formalised papers and Comparator challenges. A family counts as “with Lean” if it has such a page; 330 of 722 papers (46%) are formalised. Having a page does not mean the headline is formalised; Section 6 audits that.

Which families are most formally verified?

Family	Lean challenges	Papers formalised	Note
238. Optimal logarithmic mixing of the Thorp shuffle	13	10 of 12	Most checked statements; includes the headline $\Theta(\log N)$ mixing time
140. Memory–sample lower bounds for noiseless Gaussian regression	8	6 of 6	Every paper formalised
331. Asymptotic midpoint uniform convexity without asymptotically uniformly convex renorming	7	7 of 7	Largest family with every paper formalised
102. The Unique Games Conjecture and optimal approximation thresholds	5	5 of 5	Unique Games: every paper and every headline theorem formalised
003. The quasi-Riemann hypothesis	4	2 of 3	Quasi-Riemann: ζ , all Dirichlet L-functions, Hecke case and Siegel-zero gap

By sheer volume of Lean code, families 266 (mutually unbiased bases) and 189 (cycle–clique Ramsey numbers) lead with about 2.8 million lines each. Both are computation-heavy proofs in which Lean re-checks enormous numbers of cases. Note that family 266's headline is *not* what its Lean proves (Section 6).

Other facts worth knowing

- **Dates.** Manuscripts are dated between 10 September and 6 October 2026. 370 are dated 23–24 September alone and 112 are dated 5 October. These look like write-up dates rather than discovery dates.
- **Two flagship results came from a different process.** The README says the zero-free region for the Riemann zeta function and the Hodge conjecture for CM abelian varieties were exceptions to the standard procedure, and that the write-up of the 11/12 zero-free region was edited by humans. It does not say what the different procedure was.
- **Missing numbers.** Families are numbered up to 377, but 045, 061, 070, 123 and 163 do not exist. It looks as if five results were withdrawn before release; the repository does not say.
- **Reasoning summaries.** Abridged “chain of thought” summaries are included for 10 results, among them π , the Mahler conjectures, free group factors and Vlasov–Maxwell (see `reasoning_traces/`).
- **OpenAI's top-level Lean catalogue is incomplete.** `lean/formalization.yaml` lists only 178 of the 405 Comparator challenges and 162 of the 330 formalised papers. The per-family pages in `lean/docs` are complete and include an honest “scope” note on what each formalisation does and does not cover.
- **Versioning.** OpenAI says corrections will be released as new versions, and that earlier versions will remain available.

3 What “verified” means here

Everything in this guide depends on one distinction: a *claim* versus a *machine-checked claim*. Here is how that works from first principles.

What a proof assistant does

A normal maths paper is checked by people. A referee reads the argument and judges whether each step follows, which is slow and can miss errors in long proofs. A **proof assistant** such as **Lean** is a programming language in which definitions, theorem statements and proofs are written so precisely that a small, trusted program (the *kernel*) can check every logical step. If the kernel accepts a proof, the statement follows from the axioms, and no human judgement about the proof is needed. **Mathlib** is the shared Lean library of standard mathematics (real numbers, complex analysis, the zeta function and so on), roughly two million lines written by the community.

What still needs a human is the **statement**: does the formal theorem say what the paper claims? If a definition is subtly wrong, a perfect proof can prove the wrong thing.

Three layers of trust

Layer	What it is	Who checks it	Size
1. The paper	The human-readable PDF argument	Nobody yet; these are unrefereed preprints	722 papers
2. The Lean statement (“challenge”)	The theorem written using Mathlib’s standard definitions, with the proof left blank	Must be audited by a human against the paper	1 to about 250 lines each
3. The Lean proof (“solution”)	The complete formal proof	A computer: the Lean kernel plus the Comparator tool	25.9M lines in total

The tool **Comparator** compiles the proof and confirms that the proved theorem has *exactly* the statement in the challenge file. It also confirms that the proof uses only the three axioms behind all of Mathlib: propositional extensionality, quotient soundness and the axiom of choice. As a result, you do not have to trust 25.9 million lines. You only have to read the short statement.

Three examples of what you would actually audit

The Riemann zeta function has no zeros with real part above $7/8$ (family 003). This uses Mathlib’s own `riemannZeta`, so nothing was redefined:

```
theorem riemannZeta_ne_zero_of_seven_eighths_lt_re
  {s : ℂ} (hs : (7 / 8 : ℝ) < s.re) : riemannZeta s ≠ 0
```

The plane cannot be coloured with five colours (family 158). Any map from the plane (here $\mathbb{C}$) to five colours puts some pair of points at distance exactly 1 in the same colour:

```
def ProperColoring (colorCount : ℕ) (coloring : ℂ → Fin colorCount) : Prop :=
  ∀ point otherPoint : ℂ, ‖point - otherPoint‖ = 1 → coloring point ≠ coloring otherPoint

theorem no_proper_five_coloring :
  ¬ ∃ coloring : ℂ → Fin 5, ProperColoring 5 coloring
```

The symmetric Mahler conjecture (family 087). For every compact, convex, origin-symmetric K in $\mathbb{R}^n$ with nonempty interior, $\text{vol}(K) \cdot \text{vol}(K^\circ) \geq 4^n/n!$:

```
theorem symmetric_mahler {n : ℕ} (hn : 1 ≤ n)
  {K : Set (Fin n → ℝ)} (hK : IsCompact K) (hconv : Convex ℝ K)
  (hsym : ∀ x ∈ K, -x ∈ K) (hint : (interior K).Nonempty) :
  (4:ℝ)^n / (Nat.factorial n:ℝ) ≤
    (volume K).toReal * (volume (coordinatePolar K)).toReal
```

What I checked in the Lean code (re-run for this edition)

- **All 405 Comparator configurations** permit exactly the three standard axioms and nothing else. Every one of the 507 named theorems is stated in its challenge file and declared in the solution code, and all 45 custom definitions are present. One configuration also switches on a second, independent kernel checker (nanoda).
- **A comment-aware scan of all 121,735 Lean files (25,911,445 lines)**, ignoring comments and strings, found 0 sorry, 0 admit, 0 axiom declarations, 0 native_decide, 0 unsafe/implemented_by/extern and no option that skips the kernel. These are the usual ways a Lean proof can cheat. The scan also flagged 30 opaque definitions, 14 custom macros or tactics and 11 environment tweaks; I inspected them. They are sealed definitions with real values, syntax helpers, and search-index or cache settings, none of which can bypass the kernel (and Comparator re-checks every proof in a fresh kernel anyway).
- A dependency called `iut` (presumably Mochizuki's disputed inter-universal Teichmüller theory) is listed in the build file, but **nothing imports it**.
- **Every challenge is accounted for**. All 405 are linked from OpenAI's per-family pages (eight are shared by two families). An independent statement-by-statement review of all of them agreed with those pages in every case.
- **9 challenges define their own concepts** rather than using Mathlib's (for example `k-server`, Naimark's problem, Rokhlin mixing, the interpolated free group factors). Statements like these need the most careful human audit.

Caveats

- I did **not** compile the library; at 25.9 million lines it is far too large for this environment. In this guide, "Lean-checked" means a clean formal proof is *present*, not that I re-ran it.
- OpenAI's catalogue says the formal proofs were written by AI agents ("automation: agent"), that the scope is "partial progress", and that review status is "unchecked".
- **A challenge's name can oversell it**. "PlanarAndersonSpectrum" proves only that a spectrum is an interval, not localisation; "MUBSix" proves at most five bases, not three. Always read the statement or the scope note in `lean/docs`. Section 6 does this for every formalised family.
- A formal proof establishes that a statement is true. It does not make the result understood.

4 Headline results

There are far too many families to cover individually (Appendix A lists them all), so this section picks the results a mathematician would recognise at once. Each card says what is claimed, explains the background from first principles, says why it matters, and states exactly what is formally verified. The wording of each claim follows OpenAI's catalogue, and the badges follow the gap audit in Section 6.

At a glance

#	Claim (click to jump to its card)	Area	Status
003	$\zeta(s) \neq 0$ for $\text{Re } s > 7/8$ (quasi-Riemann hypothesis); no Landau–Siegel zeros	Number theory	✓ LEAN-CHECKED
017	The irrationality exponent of π is 2	Number theory	✓ LEAN-CHECKED
002	Full BSD formula in Selmer corank 0 and 1; Goldfeld's conjecture	Number theory	○ PAPER ONLY
004	Hilbert's tenth problem over $\mathbb{Q}$ is undecidable	Number theory	○ PAPER ONLY
005	Catalan's constant is irrational	Number theory	✓ LEAN-CHECKED
029	Artin's primitive root conjecture (infinitude), every base	Number theory	○ PAPER ONLY
030	Modularity of elliptic curves over imaginary quadratic fields	Number theory	○ PAPER ONLY
028	Gaussian moat: no bounded-step walk to infinity on Gaussian primes	Number theory	✓ LEAN-CHECKED
032	Hodge conjecture for CM abelian varieties	Algebraic geometry	○ PAPER ONLY
087	Mahler conjectures (symmetric and non-symmetric)	Convex geometry	✓ LEAN-CHECKED
102	The Unique Games Conjecture	Theoretical CS	✓ LEAN-CHECKED
107	Matrix multiplication exponent $\omega \leq 9/4$	Theoretical CS	✓ LEAN-CHECKED
103	L = RL = BPL (randomness does not help log-space)	Theoretical CS	○ PAPER ONLY
138	Subset Sum, integer multiplication and the DFT, all faster	Theoretical CS	🟡 PARTLY LEAN
158	The plane cannot be coloured with five colours	Combinatorics	✓ LEAN-CHECKED
156	Borsuk's conjecture fails in dimension 9	Combinatorics	✓ LEAN-CHECKED
155	Periodic tiling conjecture fails in dimension 3	Combinatorics	✓ LEAN-CHECKED
248	Thompson's group F is nonamenable	Group theory	✓ LEAN-CHECKED
196	Kaplansky's zero-divisor conjecture is false	Algebra	✓ LEAN-CHECKED
287	All free group factors are isomorphic	Operator algebras	✓ LEAN-CHECKED
288	Kadison's similarity problem	Operator algebras	✓ LEAN-CHECKED
246	Cannon's conjecture	Group theory	✓ LEAN-CHECKED
304	Hilbert–Smith conjecture in every dimension	Topology	○ PAPER ONLY
362	Global smoothness for relativistic Vlasov–Maxwell	PDE	✓ LEAN-CHECKED

Number theory

FAMILY 003 · NUMBER THEORY · 2 OF 3 PAPERS FORMALISED · 4 LEAN CHALLENGES

A zero-free half-plane for ζ : the quasi-Riemann hypothesis

✓ LEAN-CHECKED

The claim. Every Dirichlet L-function, including the Riemann zeta function $\zeta(s)$, has no zeros with real part greater than $7/8$. The same half-plane is zero-free for finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$. A companion paper reaches $\operatorname{Re} s > 11/12$ by a different route, and another rules out Landau–Siegel zeros.

From first principles. The zeta function is $\zeta(s) = 1 + 1/2^s + 1/3^s + \dots$. Euler noticed that it factors over the primes, $\zeta(s) = \prod_p 1/(1 - p^{-s})$, so ζ encodes every prime at once. Riemann showed that the zeros of ζ control how evenly the primes are spread. Concretely, if no zero has real part bigger than θ , then the number of primes up to x , written $\pi(x)$, differs from its smooth approximation $\operatorname{Li}(x) = \int_2^x dt / \log t$ by at most about x^θ . The Riemann Hypothesis says $\theta = 1/2$. Until now nobody could prove *any* θ below 1. The known zero-free regions hug the line $\operatorname{Re} s = 1$ and get thinner as you go up (their width shrinks roughly like $1/\log t$).

Why it matters. A fixed zero-free strip immediately gives $\pi(x) = \operatorname{Li}(x) + O(x^{7/8+\epsilon})$. That would be the first power-saving error term in the prime number theorem, 130 years after the theorem was proved. Separately, possible “exceptional” Siegel zeros near $s = 1$ have blocked effective bounds on class numbers and on primes in arithmetic progressions since the 1930s.

Verification. All four main statements are formalised: $\zeta(s) \neq 0$ for $\operatorname{Re} s > 7/8$ (one line, Section 3), the same for every Dirichlet L-function and for finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$, and the Landau–Siegel statement: there is an absolute $c > 0$ such that no real primitive Dirichlet character $\chi \bmod q$ has a real zero β with $1 - \beta < c / \log q$. Only the alternate $11/12$ proof and later applications are outside. The README says this result was produced outside the standard pipeline.

FAMILY 017 · NUMBER THEORY · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

The irrationality exponent of π is exactly 2

✓ LEAN-CHECKED

The claim. For every $\epsilon > 0$ and every large enough q , every fraction p/q satisfies $|\pi - p/q| \geq q^{-2-\epsilon}$. As a consequence, the Flint–Hills series $\sum 1/(n^3 \sin^2 n)$ converges.

From first principles. Any irrational number x can be approximated by fractions; the question is how well, relative to the size of the denominator. Dirichlet showed that every irrational x has infinitely many fractions with $|x - p/q| < 1/q^2$; continued fractions produce them. The *irrationality exponent* $\mu(x)$ is the largest power μ for which $|x - p/q| < 1/q^\mu$ still has infinitely many solutions, so $\mu(x) \geq 2$ always. Almost every real number has $\mu = 2$ exactly, but proving this for a specific constant is notoriously hard. For π the best proven bound was $\mu(\pi) \leq 7.103$ (Zeilberger–Zudilin, 2020).

Why it matters. $\mu(\pi) = 2$ says that π is, in this precise sense, as hard to approximate as a “typical” number. The Flint–Hills series was a famous test case, since its convergence was known to follow from $\mu(\pi) < 5/2$.

Verification. Formalised: the Lean statement says the irrationality exponent of π is exactly 2. A reasoning summary is included; see Section 7.

FAMILY 002 + 006 · NUMBER THEORY · 0 OF 5 PAPERS FORMALISED · 0 LEAN CHALLENGES

Birch–Swinnerton-Dyer in ranks 0 and 1, and Goldfeld’s conjecture

○ PAPER ONLY

The claim. The full BSD leading-term formula, including finiteness of the Tate–Shafarevich group $\mathbb{W}$, holds for every elliptic curve over $\mathbb{Q}$ whose q -power Selmer group has corank 0 or 1 for some prime q . Goldfeld’s conjecture also holds: among the quadratic twists of any elliptic curve, ranks 0 and 1 each occur with density $1/2$. Together these give full BSD for 100% of quadratic twists.

From first principles. An elliptic curve is an equation such as $y^2 = x^3 + ax + b$. Its rational points form a finitely generated group (Mordell), and the *rank* r counts the independent points of infinite order. From counting solutions modulo each prime one also builds an analytic function, the L-function $L(E, s)$. The Birch–Swinnerton-Dyer

conjecture (a Millennium Prize problem) says that r equals the order of vanishing of $L(E, s)$ at $s = 1$. It also predicts the exact leading coefficient in terms of arithmetic invariants, including the mysterious group $\mathbb{W}$. Gross–Zagier and Kolyvagin (1980s) proved the rank equality when the analytic rank is 0 or 1, but the exact formula stayed open.

Why it matters. This is essentially the whole BSD conjecture in the ranks that occur most often, and Goldfeld's conjecture says those ranks are typical.

Verification. Paper only.

FAMILY 004 · NUMBER THEORY · 0 OF 2 PAPERS FORMALISED · 0 LEAN CHALLENGES

Hilbert's tenth problem over $\mathbb{Q}$

☐ PAPER ONLY

The claim. No algorithm can decide whether a polynomial equation with integer coefficients has a solution in rational numbers.

From first principles. In 1900 Hilbert asked for a procedure to decide whether a polynomial equation (for example $x^3 + y^3 = 29z^3$) has a solution in whole numbers. Matiyasevich (1970), building on Davis, Putnam and Robinson, proved that no such algorithm exists: equations over the integers can encode arbitrary computer programs. Over the rationals the question stayed open for more than fifty years, because the tricks that force a variable to be an integer break down for fractions.

Why it matters. It settles one of the most-studied open problems between logic and number theory.

Verification. Paper only.

FAMILY 005 · NUMBER THEORY · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

Catalan's constant is irrational

☒ LEAN-CHECKED

The claim. $G = 1 - 1/3^2 + 1/5^2 - 1/7^2 + \dots \approx 0.91597$ is not a fraction.

From first principles. Proving that a specific constant is irrational is often extremely hard. $\zeta(3)$ was only done by Apéry in 1978, and $\zeta(5)$ is still open. Catalan's constant is the alternating cousin of $\pi^2/8 = 1 + 1/3^2 + 1/5^2 + \dots$, and its irrationality has been a standard open problem for well over a century.

Verification. Formalised: the Lean statement is literally " $\sum (-1)^j/(2j+1)^2$ is irrational". The reasoning summary for the π result cites the determinant technique of this paper as an inspiration.

FAMILY 029 · NUMBER THEORY · 0 OF 2 PAPERS FORMALISED · 0 LEAN CHALLENGES

Artin's primitive root conjecture (infinitude), for every base

☐ PAPER ONLY

The claim. Every integer a that is not -1 and not a perfect square is a primitive root modulo infinitely many primes. For each such a , at least $c_a x/(\log x)^2$ primes in $(x, 2x)$ qualify.

From first principles. Modulo a prime p , the nonzero remainders $1, \dots, p-1$ form a cyclic group under multiplication. A number a is a *primitive root* mod p if its powers $a, a^2, a^3, \dots$ run through all of them; for example, the powers of 3 modulo 7 are 3, 2, 6, 4, 5, 1. Artin (1927) conjectured that every a other than -1 and perfect squares is a primitive root for infinitely many primes. Hooley proved this in 1967 assuming a generalised Riemann Hypothesis. Unconditionally, Heath-Brown (1986) showed that at most two prime bases can fail, but could not say which, so not even $a = 2$ was known.

Verification. Paper only.

FAMILY 030 · NUMBER THEORY · 0 OF 1 PAPER FORMALISED · 0 LEAN CHALLENGES

Modularity of elliptic curves over imaginary quadratic fields

☐ PAPER ONLY

The claim. Every elliptic curve over every imaginary quadratic field, such as $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-5})$, is modular, with matching local data at every place.

From first principles. The modularity theorem of Wiles, Taylor–Wiles and Breuil–Conrad–Diamond–Taylor says that every elliptic curve over $\mathbb{Q}$ comes from a modular form; this is what proved Fermat’s Last Theorem. A central programme is to extend this to other number fields. Over imaginary quadratic fields the relevant automorphic forms live on hyperbolic 3-space and do not come from algebraic geometry in the usual way, which makes the problem much harder. Only partial results were known.

Verification. Paper only.

FAMILY 028 · NUMBER THEORY · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

The Gaussian moat conjecture

✓ LEAN-CHECKED

The claim. There is no infinite walk through distinct Gaussian primes with steps of bounded length. More strongly, for each step bound D , every cluster of Gaussian primes linked by steps of length at most D is finite, with a size bound that depends only on D .

From first principles. Gaussian integers are complex numbers $a + bi$ with a and b whole numbers, and Gaussian primes are those that cannot be factored further (for example 3, or $1 + 2i$). The question, posed in 1962, asks whether you can walk from the origin to infinity stepping only on Gaussian primes with steps no longer than some fixed length. Computer searches had found ever-wider prime-free “moats”, but there was no proof. On the ordinary number line the answer is easy, because prime gaps grow without bound.

Verification. Formalised, including the uniform bound on cluster sizes.

Algebraic geometry

FAMILY 032 · ALGEBRAIC GEOMETRY · 0 OF 8 PAPERS FORMALISED · 0 LEAN CHALLENGES

The Hodge conjecture for CM abelian varieties

○ PAPER ONLY

The claim. The rational Hodge conjecture holds for every complex abelian variety with complex multiplication, in every dimension and codimension. Companion papers prove it for arbitrary products of K3 surfaces. Via Milne’s theorems this gives the Tate conjecture for all abelian varieties over finite fields.

From first principles. A smooth complex projective variety is a shape cut out by polynomial equations. Its cohomology (a way of counting holes of each dimension) contains special classes called *Hodge classes*, which look algebraic from the point of view of analysis. Every subvariety produces a Hodge class. The Hodge conjecture, a Millennium Prize problem, says that with rational coefficients every Hodge class arises this way. Abelian varieties are higher-dimensional analogues of elliptic curves, and those with complex multiplication (CM) have extra symmetry that creates many “exotic” Hodge classes. They have long been a test case where the conjecture was open.

Why it matters. This would be one of the largest known cases of a Millennium problem, with consequences in characteristic p as well.

Verification. Paper only. The README says this result was produced outside the standard pipeline.

Convex geometry

FAMILY 087 · CONVEX GEOMETRY · 3 OF 3 PAPERS FORMALISED · 4 LEAN CHALLENGES

The Mahler conjectures

✓ LEAN-CHECKED

The claim. For every convex body K in $\mathbb{R}^n$ that is symmetric about the origin, $\text{vol}(K) \cdot \text{vol}(K^\circ) \geq 4^n/n!$. Equality holds exactly for Hanner polytopes, such as the cube. The non-symmetric version holds too, with simplices as the minimisers.

From first principles. A convex body is a bounded convex shape with interior, like a ball, a cube or an ellipsoid. Its *polar* is $K^\circ = \{y : x \cdot y \leq 1 \text{ for every } x \in K\}$. Stretching K squashes K° and vice versa, so the product $\text{vol}(K) \cdot \text{vol}(K^\circ)$ does not change under any invertible linear map; it measures shape alone. The ball maximises it (the Blaschke–Santaló inequality). Mahler (1939) conjectured that the cube minimises it among symmetric bodies, which gives the bound $4^n/n!$. This was known only in dimensions 2 and 3, and dimension 3 was settled only in 2020.

Why it matters. It is a basic inequality of convex geometry, with links to the geometry of numbers, functional analysis and symplectic geometry. The family also computes the Gromov width of $K \times K^\circ$.

Verification. Both cases are formalised for every n : the symmetric inequality with its equality cases (Hanner bodies; statement in Section 3), and the general inequality with equality exactly for simplices. The functional versions are outside. A reasoning summary is included.

Theoretical computer science

FAMILY 102 · THEORETICAL CS · 5 OF 5 PAPERS FORMALISED · 5 LEAN CHALLENGES

The Unique Games Conjecture

✓ LEAN-CHECKED

The claim. Khot's Unique Games Conjecture is true. Separate direct proofs show it is NP-hard to beat the Goemans–Williamson ratio for Max-Cut, and to approximate Vertex Cover below a factor of 2.

From first principles. Many optimisation problems are NP-hard to solve exactly, so we ask how close an efficient algorithm can get. A *Unique Games* instance is a graph, a set of k labels, and on each edge a permutation π saying “if this end gets label a , the other end must get $\pi(a)$ ”. If every constraint can be satisfied, a solution is easy to find: label one vertex and propagate. Khot (2002) conjectured that the approximate version is hard. For any $\varepsilon, \delta > 0$, it should be NP-hard to tell apart instances where at least $1 - \varepsilon$ of the constraints can be satisfied from instances where at most δ can.

Why it matters. If true, many known algorithms are exactly optimal (unless $P = NP$): the 0.878 ratio of Goemans–Williamson for Max-Cut, factor 2 for Vertex Cover, and, by Raghavendra's theorem, the basic semidefinite-programming algorithm for every constraint satisfaction problem. It is arguably the central open problem in hardness of approximation.

Verification. The conjecture itself is formalised, as a polynomial-time Turing-machine reduction from 3-SAT to Unique Games with completeness at least $1 - \varepsilon$ and soundness at most δ , for all $0 < \varepsilon, \delta < 1/2$. The Max-Cut and Vertex Cover theorems are formalised too. The statement uses about 240 lines of definitions, so it deserves close human audit. A reasoning summary covers a related paper.

FAMILY 107 · THEORETICAL CS · 2 OF 3 PAPERS FORMALISED · 2 LEAN CHALLENGES

Matrix multiplication with exponent $\omega \leq 9/4$

✓ LEAN-CHECKED

The claim. Two $n \times n$ complex matrices can be multiplied with $O(n^{9/4+\varepsilon})$ arithmetic operations, so $\omega \leq 2.25$. The dual exponent satisfies $\alpha > 0.465$. Over every field $\omega < 2.37106$, and outside finitely many positive characteristics $\omega < 2.258$.

From first principles. Multiplying two $n \times n$ matrices the schoolbook way takes about n^3 multiplications: each of the n^2 entries of the answer is a sum of n products. In 1969 Strassen found how to multiply 2×2 matrices with 7

multiplications instead of 8; applied recursively, this gives $n^{2.807}$. The exponent ω is the best achievable power, and $\omega \geq 2$ because the answer alone has n^2 entries. Decades of increasingly elaborate methods lowered the bound to about 2.371 (2024). Between 1990 and 2024 it moved by only about 0.004.

Why it matters. Matrix multiplication sits underneath linear algebra, graph algorithms and much of complexity theory, so ω appears in the running time of hundreds of algorithms. A drop from 2.371 to 2.25 would be by far the largest improvement in more than 35 years.

Verification. The headline bounds are formalised: $\omega \leq 9/4$ over $\mathbb{C}$, $\alpha > 0.465$, and $\omega < 2.37106$ over every field. The challenge defines ω in the textbook way: straight-line arithmetic programs, where each $+$, $-$ and $\times$ costs one step.

Minor gap vs the title. Lean proves: $\omega(\mathbb{C}) \leq 9/4$, $\alpha > 0.465$, $\omega(\mathbb{C}; 1, 0.709, 1) < 2.092$, and $\omega(F) < 2.371054886006746$ for every field F . Not formalised: The $\omega < 2.258$ bound for all but finitely many positive characteristics is not formalized.

FAMILY 103 · THEORETICAL CS · 0 OF 1 PAPER FORMALISED · 0 LEAN CHALLENGES

Derandomising logarithmic space: $L = RL = BPL$

 PAPER ONLY

The claim. Every problem a randomised algorithm can solve with $O(\log n)$ memory, with one- or two-sided bounded error, can be solved deterministically with $O(\log n)$ memory. The paper gives an explicit compiler.

From first principles. The resource here is memory rather than time. L is the class of problems a deterministic machine can solve with memory proportional to $\log n$, which is enough for a few pointers into the input. RL and BPL let the machine flip coins, with one-sided or two-sided error. Does randomness ever help? For space, the evidence said no, but the best unconditional result for decades was $BPL \subseteq \text{SPACE}(\log^{3/2} n)$ (Saks–Zhou, 1999). Reingold's 2005 proof that undirected connectivity is in L was a famous special case.

Verification. Paper only.

FAMILY 138 + 109 + 130 · THEORETICAL CS · 1 OF 5 PAPERS FORMALISED · 1 LEAN CHALLENGE

Faster basic algorithms: Subset Sum, integer multiplication, Fourier transform

 PARTLY LEAN

The claim. Three classic running-time barriers are broken. Subset Sum runs in $O(2^{0.49n})$ time (family 138). Integers multiply in $O(n (\log n)^{1-\kappa})$ time with $\kappa = 2^{-182}$ (family 109). The discrete Fourier transform uses $O(n (\log n)^{1-\delta})$ operations with $\delta = 10^{-13}$ (family 130).

From first principles. *Subset Sum* asks whether some of n given integers add up to a target. Trying all subsets takes 2^n steps. “Meet in the middle” (Horowitz–Sahni, 1974) cuts this to about $2^{n/2}$, and that exponent had not been beaten for worst-case inputs in 50 years. For *integer multiplication*, Harvey and van der Hoeven (2019) reached $n \log n$, long conjectured to be optimal; the new bound beats it on multitape Turing machines, although a gain of 2^{-182} in the exponent of a logarithm is invisible in practice. The *fast Fourier transform* has run in $n \log n$ since 1965; the claim beats that in an exact-arithmetic model.

Verification. Subset Sum and integer multiplication have no formal proof. For the Fourier transform, Lean proves a weaker statement than the title: circuits beating $c \cdot n \cdot \log n$ exist for an unbounded sequence of lengths n , not for every n .

Combinatorics

FAMILY 158 · COMBINATORICS · 1 OF 1 PAPER FORMALISED · 2 LEAN CHALLENGES

The plane cannot be coloured with five colours

✓ LEAN-CHECKED

The claim. Every colouring of the plane with five colours has two points at distance exactly 1 with the same colour. There are no restrictions on what the colour classes look like.

From first principles. Colour every point of the plane so that any two points exactly one unit apart get different colours. What is the smallest number of colours that works? This is the “chromatic number of the plane”, from the Hadwiger–Nelson problem (1950). A tiling by hexagons of diameter slightly under 1 shows that 7 colours suffice. Lower bounds come from finite unit-distance graphs that cannot be coloured with few colours. For decades the best lower bound was 4, until de Grey (2018) found a graph that needs 5.

Why it matters. The answer is now pinned to 6 or 7.

Verification. Formalised, together with the classical fact that seven colours suffice. The Lean statement is three lines; see Section 3.

FAMILY 156 · COMBINATORICS · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

Borsuk’s conjecture fails in dimension 9

✓ LEAN-CHECKED

The claim. There is a compact set in $\mathbb{R}^9$ that cannot be split into 10 pieces of smaller diameter.

From first principles. The diameter of a set is the largest distance between two of its points. Borsuk (1933) asked whether every bounded set in $\mathbb{R}^n$ can be split into $n + 1$ pieces, each of strictly smaller diameter; a ball shows that $n + 1$ cannot be lowered. The answer is yes in dimensions 1, 2 and 3. Kahn and Kalai (1993) showed it fails in dimension 1325, and the smallest dimension with a known counterexample was 64 (Jenrich–Brouwer, 2014).

Why it matters. The unknown range shrinks to dimensions 4 to 8. The example is built from rank-one orthogonal projectors onto lines in $\mathbb{R}^4$.

Verification. Formalised.

FAMILY 155 · COMBINATORICS · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

The periodic tiling conjecture fails in dimension 3

✓ LEAN-CHECKED

The claim. There is a finite set of cells in $\mathbb{Z}^3$ whose translates tile space, but never periodically. Thickening the cells to unit cubes gives the same result in $\mathbb{R}^3$.

From first principles. Suppose copies of a single shape, moved only by translations (no rotations), can fill space without overlaps. Must there then also be a *periodic* way to do it, one that repeats in every direction? This is true in dimension 1, and for the lattice $\mathbb{Z}^2$ (Bhattacharya, 2016). Greenfeld and Tao (2022) disproved it in some very high dimension.

Why it matters. Since the conjecture holds in $\mathbb{Z}^2$, dimension 3 is the smallest possible counterexample.

Verification. Formalised; Lean also proves that 3 is the smallest dimension where this can happen.

Groups, algebra and operator algebras

FAMILY 248 · GROUP THEORY · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

Thompson's group F is nonamenable

✓ LEAN-CHECKED

The claim. Thompson's group F is not amenable.

From first principles. A group G is *amenable* if every bounded function on G can be given an “average” that does not change when the function is shifted by group elements. Finite groups and abelian groups are amenable; the free group on two generators is not. Thompson's group F consists of the piecewise-linear bijections of $[0, 1]$ whose breakpoints are dyadic rationals and whose slopes are powers of 2. It sits right on the boundary: it contains no free subgroup, the usual obstruction, yet no averaging had ever been found.

Why it matters. The question has been asked since about 1979 and is among the best-known problems in group theory. Several claimed proofs, in both directions, were later found to be wrong.

Verification. Formalised. The challenge defines F concretely and states that no invariant mean exists.

FAMILY 196 · ALGEBRA · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

A counterexample to Kaplansky's zero-divisor conjecture

✓ LEAN-CHECKED

The claim. There is a finitely presented torsion-free group G whose group algebra $F_2[G]$ has nonzero zero divisors.

From first principles. The group algebra of G consists of finite formal combinations of group elements, multiplied using the group law. If G has an element g of finite order n, then $(1 - g)(1 + g + \dots + g^{n-1}) = 0$, so zero divisors appear. Kaplansky (1950s) conjectured that torsion-free groups never have them. Gardam (2021) disproved the related unit conjecture; the zero-divisor conjecture remained open.

Verification. Formalised.

FAMILY 287 · OPERATOR ALGEBRAS · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

All free group factors are isomorphic

✓ LEAN-CHECKED

The claim. $L(F_2) \cong L(F_3)$. Hence all free group factors $L(F_n)$, $n \geq 2$, and the interpolated ones, including $L(F_\infty)$, are isomorphic.

From first principles. From any group G one builds a von Neumann algebra $L(G)$: the closure of the algebra of operators that G generates when it acts on $\ell^2(G)$. For the free groups F_n on n generators these algebras are “factors” with a natural trace. Whether $L(F_2)$ and $L(F_3)$ are isomorphic is one of the most famous problems in operator algebras. In the 1990s Dykema and Rădulescu showed that either all $L(F_n)$ with $n \geq 2$ are isomorphic, or all are pairwise different.

Why it matters. This settles the dichotomy in the “all isomorphic” direction. Voiculescu's free probability theory was developed largely as a tool for this question.

Verification. Formalised. The statement builds the group von Neumann algebra on $\ell^2(G)$ from scratch, defines the interpolated factors $L(F_r)$ as corners of an amplified $L(F_2)$, and proves them all isomorphic for $r > 1$, which includes $L(F_2) \cong L(F_3)$. It runs to about 225 lines, so it deserves expert audit. A reasoning summary is included.

FAMILY 288 · OPERATOR ALGEBRAS · 1 OF 1 PAPER FORMALISED · 2 LEAN CHALLENGES

Kadison's similarity problem

✓ LEAN-CHECKED

The claim. Every bounded unital homomorphism from a C^* -algebra into the bounded operators on a Hilbert space becomes a $*$ -homomorphism after conjugation by a bounded invertible operator.

From first principles. A C^* -algebra is an algebra of operators closed under taking adjoints and limits. Kadison (1955) asked whether representations that are merely bounded are always, after a change of coordinates, of the

well-behaved “self-adjoint” kind. This was known for important classes of algebras but open in general.

Verification. Formalised for arbitrary Hilbert spaces, together with a universal commutator estimate and a hyperreflexivity bound.

Topology, geometry and PDE

FAMILY 246 · GROUP THEORY · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

Cannon’s conjecture

✓ LEAN-CHECKED

The claim. Every hyperbolic group whose boundary is a 2-sphere acts geometrically on hyperbolic 3-space. The torsion-free ones are fundamental groups of closed hyperbolic 3-manifolds.

From first principles. A hyperbolic group is one whose Cayley graph looks negatively curved from far away, and it has a boundary “at infinity”. Cannon’s conjecture (1980s) says that if this boundary is a 2-sphere, the group really comes from three-dimensional hyperbolic geometry. It is a key missing piece in recognising 3-manifold groups from their large-scale geometry.

Verification. Formalised: such a group has a proper, cocompact isometric action on hyperbolic 3-space with finite kernel.

FAMILY 304 · TOPOLOGY · 0 OF 1 PAPER FORMALISED · 0 LEAN CHALLENGES

The Hilbert–Smith conjecture in every dimension

○ PAPER ONLY

The claim. Every locally compact group acting faithfully and continuously on a connected finite-dimensional manifold is a Lie group.

From first principles. Hilbert’s fifth problem asked whether continuous groups of transformations are automatically smooth (Lie groups). The main case was solved by Gleason, Montgomery and Zippin in the 1950s. The Hilbert–Smith conjecture is the remaining general form, and its crucial case is to rule out actions of the p -adic integers $\mathbb{Z}_p$. It was known only for manifolds of dimension at most 3 (Pardon, 2013).

Verification. Paper only.

FAMILY 362 · PDE · 1 OF 1 PAPER FORMALISED · 1 LEAN CHALLENGE

Global smoothness for the relativistic Vlasov–Maxwell system

✓ LEAN-CHECKED

The claim. In three dimensions, take any smooth initial data with compactly supported particle density and finite-energy fields. The solution then exists, is unique and stays smooth for all time.

From first principles. This system models a collisionless plasma: a cloud of charged particles moving near the speed of light, described by a density $f(t, x, p)$ over position and momentum, coupled to the electric and magnetic fields the particles themselves generate (Maxwell’s equations). The danger is that particles concentrate and the fields blow up. Glassey and Strauss (1986) proved global smoothness provided particle momenta stay bounded. Removing that condition was a long-standing open problem in kinetic theory.

Verification. Formalised, with no smallness, symmetry or neutrality assumption. A reasoning summary is included.

5 Physics results

About 64 families concern physics: the 25 in the catalogue's "Mathematical physics" section, the 29 in "Probability and statistical mechanics", and a dozen PDE and dynamics results on fluids, plasmas, kinetic theory and chaos. Lean pages exist for 17 of the 25 mathematical-physics families and 19 of the 29 statistical-mechanics ones. However, in many of them Lean checks a supporting or weaker statement rather than the headline. The four cards below are major physics problems whose central statement *is* Lean-checked (where the family's title claims more, the card says so), and the table after them gives the status of every physics family.

Four major physics problems with Lean proofs

FAMILY 215 · STATISTICAL MECHANICS · 1 OF 5 PAPERS FORMALISED · 1 LEAN CHALLENGE

Polyakov's conjecture: 2D spin models with $n \geq 3$ components never order ✓ LEAN-CHECKED

The claim. In the two-dimensional nearest-neighbour $O(n)$ model with $n \geq 3$, spin correlations decay exponentially at every positive temperature. The family also claims the continuum limit of the $O(3)$ model (a quantum field theory with a mass gap) and the exact mass asymptotics of the $O(4)$ model.

From first principles. Put a small arrow (a unit vector with n components) on each site of a square grid, with neighbouring arrows preferring to align at coupling strength β (inverse temperature). In two dimensions the Mermin–Wagner theorem (1966) rules out true long-range order. For $n = 2$ there is still a strange "BKT" phase in which correlations fall off only as a power law. For $n \geq 3$, Polyakov predicted in 1975 that correlations always decay exponentially, however cold the system.

Why it matters. The $O(3)$ model is the standard toy model for quantum chromodynamics: it shares asymptotic freedom and a dynamically generated mass gap, making it a two-dimensional cousin of the Yang–Mills mass-gap Millennium problem.

Verification. The exponential decay is formalised: for every $n \geq 3$ and every $\beta > 0$, $\langle \sigma_x \cdot \sigma_y \rangle \leq A \cdot e^{-m|x-y|}$ uniformly over all finite boxes. The continuum $O(3)$ theory and the $O(4)$ mass asymptotics are paper only.

Gap vs the family's title. Lean proves: Finite-volume exponential decay of spin correlations for 2D classical $O(n)$, $n \geq 3$, uniform over finite subgraphs with bounded couplings. Not formalised: $O(3)$ continuum limit/mass gap and $O(4)$ exact mass asymptotics (the title claims) not formalized; infinite-volume limit also excluded.

FAMILY 213 · STATISTICAL MECHANICS · 2 OF 2 PAPERS FORMALISED · 2 LEAN CHALLENGES

No infinite cluster at critical percolation, including three dimensions ✓ LEAN-CHECKED

The claim. At the critical probability, Bernoulli percolation has no infinite cluster on every quasi-transitive graph with $p_c < 1$, and both bond and site percolation on $\mathbb{Z}^3$ have none.

From first principles. Keep each bond of a lattice independently with probability p . Above a critical value p_c an infinite connected cluster appears. Exactly at p_c , is there one? Physically this asks whether the phase transition is continuous. The answer "no" was known in two dimensions (Kesten, 1980) and in high dimensions ($d \geq 11$), but the physically central three-dimensional case was one of the best-known open problems in probability.

Verification. Formalised, both for $\mathbb{Z}^3$ (bond and site) and in the general Benjamini–Schramm form.

FAMILY 269 · MATHEMATICAL PHYSICS · 2 OF 2 PAPERS FORMALISED · 4 LEAN CHALLENGES

The Laughlin gap in the fractional quantum Hall effect ✓ LEAN-CHECKED

The claim. For the fermionic Laughlin state at filling $1/3$ on the sphere, the Haldane V_1 interaction has a spectral gap that stays bounded below as the number of electrons grows, and the gap survives weak disorder.

From first principles. Electrons confined to a plane in a strong magnetic field have their kinetic energy quantised into "Landau levels". In the lowest level every state has the same kinetic energy, so the interactions alone decide the physics. At filling $1/3$, Laughlin's wavefunction $\prod (z_i - z_j)^3$ is the exact zero-energy ground state of Haldane's

model interaction. The quantised Hall plateaus seen in experiments require an energy gap above it that does not close as the system grows: Haldane's 1983 conjecture.

Verification. The gap is formalised with an explicit constant: for all large N , every antisymmetric state has energy at least $1/25$ times its squared distance from the Laughlin state. Stability under disorder is outside the formal statement.

Minor gap vs the title. Lean proves: Uniform V1 spectral gap ($1/100$, and $1/25$ distance bound) above the Laughlin state at flux $3(N-1)$; Fock-space and planar inequalities. Not formalised: Stability of the gap and uniqueness of the ground state under weak projected scalar potentials (title's second claim) not formalized.

FAMILY 271 · MATHEMATICAL PHYSICS · 1 OF 4 PAPERS FORMALISED · 1 LEAN CHALLENGE

Spontaneous magnetisation of the quantum Heisenberg ferromagnet

✓ LEAN-CHECKED

The claim. The nearest-neighbour quantum Heisenberg ferromagnet on $\mathbb{Z}^d$, $d \geq 3$, is magnetised at low temperature for every spin S ; the family also proves Bloch's $T^{3/2}$ law with its exact coefficient.

From first principles. Quantum spins on a lattice that prefer to align should, below some temperature, pick a common direction spontaneously. For the classical version this was proved in 1976 and for the quantum antiferromagnet in 1978, both with a technique called reflection positivity. That technique fails for the quantum ferromagnet, which is why this case had stayed open; it appears on Barry Simon's 1984 list of open problems in mathematical physics, if I recall that list correctly.

Verification. Spontaneous magnetisation is formalised for every spin S and every $d \geq 3$, as a translation-invariant KMS equilibrium state. Bloch's law is paper only. A reasoning summary is included.

Gap vs the family's title. Lean proves: Spontaneous magnetization: nearest-neighbor Heisenberg ferromagnet on $\mathbb{Z}^d$, $d \geq 3$, any spin, has low-temperature KMS state with magnetization $\geq S/4$. Not formalised: Bloch's $T^{3/2}$ law with exact coefficient (the headline) and the first lattice correction are not formalized.

Every physics family at a glance

"What Lean checks" is based on OpenAI's scope note for each family, compared with the title by the gap audit in Section 6.

#	Result	Status	What Lean checks
211	Geometry, diffusion, and spectra of random planar maps	ⓘ PARTLY LEAN	Only $q > 4$: rescaled critical finite FK maps converge in Gromov-Hausdorff-Prokhorov distribution to the Brownian continuum random tree. Not formalised: LQG convergence for $0 < q \leq 4$, the surface-to-tree transition, Liouville Brownian motion and eigenvalue/heat-trace convergence are unformalized.
212	No bigeodesics and smooth limit shapes in planar first-passage percolation	ⓘ PARTLY LEAN	Differentiability of the time-constant norm ($\mathbb{C}^1$ unit sphere) for every Gamma edge-weight law, including exponential with unique supporting lines. Not formalised: Nonexistence of bigeodesics (title's headline) is not formalized; strict convexity of the exponential limit shape explicitly outside.
213	No infinite critical clusters on quasi-transitive graphs	✓ LEAN-CHECKED	The headline claim.
214	The Benjamini-Schramm nonuniqueness conjecture	✓ LEAN-CHECKED	The headline claim.
215	Massive continuum limits and exact mass asymptotics for planar $O(n)$ models	ⓘ PARTLY LEAN	Finite-volume exponential decay of spin correlations for 2D classical $O(n)$, $n \geq 3$, uniform over finite subgraphs with bounded couplings. Not formalised: $O(3)$ continuum limit/mass gap and $O(4)$ exact mass asymptotics (the title claims) not formalized; infinite-volume limit also excluded.
216	Critical logarithmic corrections and BKT scaling for the planar XY model	○ PAPER ONLY	No formal proof.

#	Result	Status	What Lean checks
217	The low-temperature Sherrington–Kirkpatrick fluctuation law	PAPER ONLY	No formal proof.
218	Conformal universality for weakly interacting and disordered Ising models	PARTLY LEAN	Finite-graph lemma: Ising conditional likelihood ratios with pinned spins bounded by $\exp(+4 \operatorname{artanh} q_0)$, q_0 an FK connection probability. Not formalised: All universality results (spin, energy, SLE_3 interface limits, disordered case) are unformalized; only a supporting comparison inequality is.
219	GOE universality for random regular graphs with weak disorder	PAPER ONLY	No formal proof.
220	Directional zero–one laws and ballisticity in random environments	LEAN-CHECKED	0-1 law for iid strictly elliptic environments, $d \geq 3$; iid uniformly elliptic: directional transience gives positive-projection deterministic velocity, $d \geq 2$. Not formalised: The zero-one law for stationary ergodic finite-range-dependent uniformly elliptic environments (unformalized third paper) is missing.
221	The Mézard–Parisi formula for diluted spin glasses	LEAN-CHECKED	The headline claim.
222	Perceptron free energies and microscopic jamming	PARTLY LEAN	Spherical perceptron pressure formula for bounded continuous potentials; Ising perceptron only finiteness of the variational value; supporting dual formula. Not formalised: Ising perceptron free-energy convergence and Borel potentials, the bi-orthogonally invariant extension, and jamming (in title) are not formalized.
223	Conformal limits of square-lattice random-cluster interfaces	PAPER ONLY	No formal proof.
224	Critical and near-critical universality for Voronoi percolation	PAPER ONLY	No formal proof.
225	Gaussian free field limits for the balanced six-vertex model	PAPER ONLY	No formal proof.
226	The double-dimer CLE_4 conjecture in the half-plane	PAPER ONLY	No formal proof.
227	The dynamical phase transition in the Sherrington–Kirkpatrick model	PARTLY LEAN	Spectral gap for $\beta < 1$; ratio cutoff only for $\beta < 1/2$; critical mixing between $n^{2/3-\varepsilon}$ and $e^{\varepsilon n}$, $\beta > 1$ stretched-exponential barrier. Not formalised: Cutoff for $1/2 \leq \beta < 1$ and its location; critical $n^{2/3+o(1)}$ upper bound (only $e^{\varepsilon n}$ proved); critical autocorrelation limits.
228	Continuum phase transitions for radial pair potentials	LEAN-CHECKED	The headline claim.
229	Sharp three- and four-state reconstruction thresholds	PARTLY LEAN	Only the supercritical direction for the symmetric 3-state channel: reconstruction when $b^* \lambda^2 > 1$ (regular) or $d^* \lambda^2 > 1$ (Poisson trees). Not formalised: Non-reconstruction at/below the threshold (the sharpness), the four-state case and stochastic-block-model consequences are not formalized.
230	Exact Hausdorff measure for SLE	PARTLY LEAN	Positivity only: every gauge agreeing with $r^d (\log \log 1/r)^{(2-d)/2}$ near 0 gives a.s. positive measure to every positive-time segment. Not formalised: Finiteness of the gauge measure and finite expected measure in bounded disks are not formalized; exactness requires both bounds.
231	The free uniform spanning forest is a factor of IID	LEAN-CHECKED	The headline claim.
232	Gaussian fields and SLE interfaces for Lipschitz heights	PAPER ONLY	No formal proof.
233	The joint critical Ashkin–Teller scaling limit	PAPER ONLY	No formal proof.

#	Result	Status	What Lean checks
234	All-temperature pressure for orthogonally invariant Ising spin glasses	✓ LEAN-CHECKED	The headline claim.
235	Random-SAT thresholds, sharp variance and computability	✓ LEAN-CHECKED	Thresholds α_k for every $k \geq 3$; $\text{Var}(H_n) = \Theta(n)$ for $k \geq 4$; computability of the 3-SAT threshold. Not formalised: For $k=3$ the hitting-time variance is only shown between $\Omega(n)$ and $O(n \log n)$, not the claimed $\Theta(n)$.
236	The factor-of-IID threshold for free Ising states on trees	✓ LEAN-CHECKED	The headline claim.
237	The three-quarter diameter exponent for honeycomb walks	ⓘ PARTLY LEAN	Supporting results: finite critical bridge masses, existence of the free-energy limit, strip-crossing mass $\sim N^{-1/4}$ and displacement moment $\sim N^{3/4}$. Not formalised: The 3/4 diameter exponent and 4/3 mass/covering exponents (the headline) are not formalized; only ingredients from 3 of 13 papers.
238	Optimal logarithmic mixing of the Thorp shuffle	✓ LEAN-CHECKED	The headline claim.
239	Sharp singularity rates for symmetric sign matrices	○ PAPER ONLY	No formal proof.
260	Spacetime Penrose inequalities and rigidity	ⓘ PARTLY LEAN	Area-controlled replacement of 3D CKS hyperboloidal ends by asymptotically flat ends, ADM masses approaching Bondi mass; Schwarzschild equality examples. Not formalised: No Penrose inequality is formalized (main inequality explicitly unformalized; only a conditional connected case retained); rigidity and charged bounds missing.
261	Localization and delocalization in the Anderson model	ⓘ PARTLY LEAN	Supporting fact: the 2D Anderson operator is a.s. bounded self-adjoint with spectrum equal to $[-4-h, 4+h]$. Not formalised: Pure-point spectral type in 2D and absolutely continuous spectrum in $d \geq 3$ (both headline claims) are not formalized.
262	Sharp one-dimensional Lieb–Thirring inequalities	✓ LEAN-CHECKED	Scalar $W \geq 0$ in $L^{Y+1/2}$: sharp one-bound-state constant bound, optimality, and attainment by $(r+1)\text{sech}^2(rx)$. Not formalised: Matrix-valued potentials (constant independent of matrix size) and the full characterization of all equality cases are not formalized.
263	The ionization and generalized ionization conjectures	ⓘ PARTLY LEAN	Thomas-Fermi asymptotics $I_m(Z)/m^{7/3} \rightarrow a_{\text{TF}}$ as $m, Z/m \rightarrow \infty$; iterated asymptotics of outer-electron radii of neutral atoms. Not formalised: The ionization conjecture ($Z+\text{CM}$ bound) and universal bounds on ionization energies and radii are not formalized; only generalized asymptotics.
264	Strong cosmic censorship near two-ended Kerr data	○ PAPER ONLY	No formal proof.
265	The two-dimensional gapped area law	○ PAPER ONLY	No formal proof.
266	Exactly three mutually unbiased bases in dimension six	ⓘ PARTLY LEAN	Weaker bound: at most five MUBs in $\mathbb{C}^6$; Fourier character-sum vanishing for non-Tao order-6 Hadamards; cube-fiber cancellation lemma. Not formalised: The bound $N(6) \leq 3$ (exclusion of four bases), the headline, is explicitly not established.
267	Positive-temperature Bose–Einstein condensation and quantum depletion	ⓘ PARTLY LEAN	Ground-state (zero-temperature) Bose-Einstein condensation in the dilute hard-sphere gas: condensate fraction $\geq c_0$ when $\rho^* a^3 < \varepsilon_0$. Not formalised: Positive-temperature condensation (the headline) and the Bogoliubov leading quantum-depletion law are not formalized.
268	The spin-one Haldane gap	○ PAPER ONLY	No formal proof.

#	Result	Status	What Lean checks
269	The Laughlin gap and stability under scalar disorder	✓ LEAN-CHECKED	Uniform V1 spectral gap (1/100, and 1/25 distance bound) above the Laughlin state at flux $3(N-1)$; Fock-space and planar inequalities. Not formalised: Stability of the gap and uniqueness of the ground state under weak projected scalar potentials (title's second claim) not formalized.
270	Threshold and positive-energy bound states of the BFSS model	○ PAPER ONLY	No formal proof.
271	Bloch's law and spontaneous ferromagnetic order	🕒 PARTLY LEAN	Spontaneous magnetization: nearest-neighbor Heisenberg ferromagnet on $\mathbb{Z}^d$, $d \geq 3$, any spin, has low-temperature KMS state with magnetization $\geq S/4$. Not formalised: Bloch's $T^{3/2}$ law with exact coefficient (the headline) and the first lattice correction are not formalized.
272	Entanglement without secret key and the PPT-square conjecture	🕒 PARTLY LEAN	PPT-square counterexample: PPT channel on 21×21 matrices whose square is not entanglement breaking; dimension-ten PPT pair composition. Not formalised: The zero-distillable-secret-key entangled state (the headline, first in title) is explicitly outside the formalization.
273	The entropy photon-number inequality	✓ LEAN-CHECKED	The headline claim.
274	Moore's parity conjecture for QAC^0	✓ LEAN-CHECKED	The headline claim.
275	QMA-hardness of continuum Coulomb energy	✓ LEAN-CHECKED	The headline claim.
276	The classical capacity of generalized amplitude damping	✓ LEAN-CHECKED	Capacity of every GAD channel for all parameters: Holevo information additive over repeated uses, equals attained scalar maximum and operational capacity. Not formalised: Additivity when tensoring with an arbitrary different partner channel (named catalogue claim) is not formalized.
277	Threshold repetition for entangled games	✓ LEAN-CHECKED	The headline claim.
278	Failure of Kohn–Sham ensemble representation	○ PAPER ONLY	No formal proof.
279	Exact quantum factoring over a fixed finite gate set	✓ LEAN-CHECKED	The headline claim.
280	Strong locality for strongly rational unitary vertex operator algebras	✓ LEAN-CHECKED	Polynomial energy bounds, CKLW strong locality, and an irreducible conformal net with covariance, vacuum and positive energy. Not formalised: Complete rationality of the net, unitarizability of simple modules, and the braided unitary tensor equivalence are not formalized.
281	QAOA optimality for the SK model	🕒 PARTLY LEAN	Parisi-value identities and finite Gaussian approximation, conditional on an admissible Parisi minimizer; full support of zero-temperature minimizers. Not formalised: Convergence of QAOA circuit energies is not asserted; results are supporting ingredients and conditional on minimizer existence; MaxCut corollary absent.
282	Scale-to-conformal enhancement in four-dimensional QFT	○ PAPER ONLY	No formal proof.
283	Polynomial-time, constant-error unitary synthesis from a Boolean oracle	○ PAPER ONLY	No formal proof.
284	The optimal randomized–quantum query exponent	○ PAPER ONLY	No formal proof.

#	Result	Status	What Lean checks
090	Universal optimality of the triangular lattice	✓ LEAN-CHECKED	Universal energy minimality for completely monotone potentials of squared distance; sharp Gaussian minorants; sharp Cohn-Elkies planar packing certificate. Not formalised: Renormalized Riesz ($0 < s < 2$) and Coulomb minimality, hence the Sandier-Serfaty and Brauchart-Hardin-Saff conjectures, are not formalized.
146	Sinai's positive-entropy conjecture for the standard map	✓ LEAN-CHECKED	The headline claim.
150	Weak mixing of irrational triangular billiards	🕒 PARTLY LEAN	Billiard flow in every nondegenerate triangle with an angle irrational to π is ergodic (flow constructed a.e., measure-preserving). Not formalised: Weak mixing (the title claim, separate paper) is not formalized; only ergodicity.
362	Global smoothness for relativistic Vlasov–Maxwell	✓ LEAN-CHECKED	The headline claim.
363	Boltzmann nonuniqueness with exact local conservation	🕒 PARTLY LEAN	Two distinct renormalized solutions with same data, local mass and total momentum conservation, energy/entropy inequalities, L^1 continuity initially. Not formalised: Exact local momentum and energy conservation (the title's distinctive claim) and global L^1 continuity are not formalized.
364	Kinetic limits and fluctuations over the Boltzmann lifespan	🕒 PAPER ONLY	No formal proof.
371	Stable blowup for a defocusing Schrödinger equation	✓ LEAN-CHECKED	The headline claim.
372	Global uniqueness in smooth isotropic elasticity	✓ LEAN-CHECKED	The headline claim.
373	Nonattainment in three-marginal Coulomb transport	✓ LEAN-CHECKED	For 3D Coulomb cost: smooth density where no Monge maps attain the Kantorovich minimum, yet infima agree. Not formalised: Extension to every inverse-power Riesz exponent in every dimension ≥ 2 is not formalized.
376	Universal computation in forced Navier–Stokes flows	✓ LEAN-CHECKED	The headline claim.

6 Gaps between titles and what Lean checks

A Lean proof only certifies the statement it proves. For every one of the formalised families I compared the family's title and catalogue claim with OpenAI's own scope note in `lean/docs`, which says explicitly what each formalisation covers and what is "outside this selected statement". Four independent reviewers classified all families, and I checked their verdicts against the Lean statements I had read myself. Each family is one of:

- **Headline formalised (146 families).** Lean proves the title's claim; at most applications or companion results are left out.
- **Minor gap (30).** The headline is formalised, but a named part of the claim (a second theorem, an equality case, a stated consequence) is not.
- **Major gap (59).** Lean proves something weaker than the title: a special case, a supporting lemma, a conditional version, or the claim only along a subsequence. These are the families where a "Lean-checked" label would mislead.
- **No Lean at all (137).** The claim rests entirely on the unrefereed paper.

Major gaps: Lean proves less than the title

#	Title	What Lean proves	Not formalised
009	Bogomolov–Pop and Milnor K-theoretic reconstruction	Only injectivity: two perfect-closure isomorphisms inducing the same Galois-data isomorphism (mod l -adic scaling) agree modulo Frobenius; $\mathrm{trdeg} \geq 2$, $\mathrm{char} \neq l$.	Existence half of Bogomolov-Pop (the actual reconstruction) and the entire Milnor K -theoretic reconstruction are not formalized.
015	Torus-packet equidistribution in prime, quartic, and sextic degrees	Weak convergence to Haar plus tightness for volume-weighted torus packets of totally real fields of fixed prime degree ≥ 5 , arbitrary local types.	Quartic Picard-packet and sextic ideal-class-packet cases (two of the three degree regimes in the title) are not formalized.
033	Campana's orbifold litaka conjecture and logarithmic subadditivity	Only the negative-fiber branch of log Kodaira additivity: if the very general fiber has log Kodaira dimension $-\infty$, so does the total space.	Campana's orbifold litaka conjecture, Popa's inequality, and the finite-dimension and negative-base additivity branches are not formalized.
049	A stable-coordinate counterexample in four variables	For $n \geq 4$, a noncoordinate F with $C[x_1..x_n]/(F) = C[y_1..y_{n-1}]$ (Abhyankar-Sathaye counterexample); commuting locally nilpotent derivations companion.	The headline stable-coordinate property (F becomes a coordinate after adjoining a variable) is not stated; only the zero fiber, not every fiber.
058	The Kollár–Pardon universal-cover conjecture	A connected bounded semialgebraic open subset of an affine algebraic set with compact proper holomorphic quotient is a bounded symmetric domain.	The Kollar-Pardon classification, quasi-projectivity criterion and C^n -covered corollary are not formalized; only a supporting symmetry result.
076	Ultraflat real Littlewood polynomials	Upper bound $\max P \leq (1+\eta)\sqrt{N}$ for all large N ; separately one family with $ P /\sqrt{N} \rightarrow 1$ in every L^p mean.	Uniform two-sided ultraflatness is not proved; merit-factor/Turyn disproof not stated (though it follows from the L^4 flatness statement).
084	The geometric case of the Erdős similarity conjecture	Only $q=1/2$: compact E in $[0,1]$ of measure $>1-\eta$ avoiding every affine copy (either sign) of $\{2^{-n}\}$.	General ratios q in $(0,1)$, the geometric-progression case claimed in the title, are not formalized; only the dyadic ratio.
095	Hyperbolicity cones without semidefinite lifts	Explicit degree-20 hyperbolic polynomial in 23 variables whose hyperbolicity cone is not spectrahedral (not a PSD slice of a linear pencil).	The headline no-semidefinite-lift claim (not a spectrahedral shadow; Projected Lax disproof) is not formalized; only non-spectrahedrality.
104	Quasipolynomial algorithms for mean-payoff, stochastic and parity games	A finite counterexample to Truffet's procedure, and a randomized (success $\geq 7/8$) quasipolynomial algorithm for mean-payoff zero-threshold winning sets.	Deterministic algorithm, exact values, optimal positional strategies, stochastic-game and parity extensions are not formalized.
114	Approximate counting of common integer polymatroid bases	FPRAS for the number of common bases of two equal-rank matroids given by independence oracles.	The polymatroid generalization in the title (integer vectors, binary capacities, rank oracles) and common-independent-set variants are not formalized.
117	Uniform sparsest cut: hardness and semidefinite gaps	Goemans-Linial integrality gap $\geq c \sqrt{(\log n)/(\log \log n)^3}$ along some sequence of instance sizes $n \rightarrow \infty$.	NP-hardness of approximating Uniform Sparsest Cut within any constant factor, the headline first claim, is not formalized.
126	Exponential semidefinite complexity of perfect matching	PSD rank of matching slack matrix and affine semidefinite lift size exceed n^C for every fixed C (superpolynomial only).	Exponential lower bound (the title claim) and the bound for every fixed shift p are explicitly not stated.
130	Fourier transforms below $n \log n$	For every $c>0$, some arbitrarily large n admit exact DFT circuits with fewer than $c n \log^2 n$ gates (subsequential).	No all-length bound and no explicit $(\log n)^{1-\delta}$ saving; only arbitrarily small normalized cost along an unbounded sequence.
143	Uniform limit-cycle bounds in Hilbert's sixteenth problem	Quintic Lienard systems $x'=y-F(x)$, $y'=-x$ with $\deg F \leq 5$ have at most two limit cycles, and two is attained.	General uniform finiteness for all planar polynomial vector fields (title claim, other paper) is not formalized; only quintic Lienard case.

#	Title	What Lean proves	Not formalised
149	Permanence for weakly reversible reaction networks	Global forward solutions exist and each trajectory stays in $[\varepsilon, 1/\varepsilon]$ with ε depending on network, rates and initial state.	Classwise permanence (common absorbing set, uniform bounds across the class) not formalized; only per-trajectory boundedness and persistence.
150	Weak mixing of irrational triangular billiards	Billiard flow in every nondegenerate triangle with an angle irrational to π is ergodic (flow constructed a.e., measure-preserving).	Weak mixing (the title claim, separate paper) is not formalized; only ergodicity.
152	Zero entropy does not guarantee a smooth positive-volume model	Ergodic invertible transformation with finite KS entropy having no smooth positive-volume model on any compact manifold.	Zero-entropy conclusion (the title claim) explicitly outside the statement; only finite entropy proved.
157	Counterexamples to the Hadwiger and Colin de Verdière conjectures	Linear List Hadwiger bound: universal C with $\chi_{\text{list}}(G) \leq C h(G)$ for all finite nonempty graphs.	The title's counterexamples to Hadwiger and fractional Colin de Verdière conjectures (other papers) are not formalized.
159	Erdős's reciprocal-sum conjecture and quasipolynomial Szemerédi bounds	Erdős's reciprocal-sum conjecture (sums of reciprocals of a set without k -term progressions).	The quasipolynomial Szemerédi bound named in the title, which is the paper's main theorem.
182	Power savings for polynomial-difference-free sets	Square-difference-free subsets of $\{1..N\}$ have size $\leq C N^{1-c}$.	Only the special case $h(m)=m^2$; general intersective polynomials and the prime-argument version are not formalized.
184	Coloring and independence in graphs with forbidden subgraphs	K_r -free graphs ($r \geq 4$) with average degree d have independent sets of size $c_r n \log d/d$.	The coloring conjecture (first headline result, other paper) is not formalized; only the independence bound.
191	A power improvement in the Heilbronn triangle problem	For an unbounded sequence of n , point sets with minimum triangle area $\geq n^{-2+\eta}$, refutes $n^{-2+\varepsilon}$ upper bound.	Construction for every sufficiently large n is not formalized; only along an unbounded sequence of sizes.
194	Lech's multiplicity conjecture	Supporting char- p lemma over a complete local domain D : Frobenius multiplicities converge to Dutta multiplicity, which bounds $e(D)$.	Lech's inequality for flat local maps in arbitrary dimension and characteristic (the main theorem) is explicitly outside the formalization.
197	Nonsofic groups and group-ring counterexamples	Direct-finiteness counterexamples: char-2 finitely presented group with odd-prime torsion; odd characteristic with non-surjunctive automaton; Determinant Conjecture counterexample.	Torsion-free group not obtained (Lean group has torsion); nonsoficity, the title's first claim, explicitly outside; spectral-integral determinant form omitted.
206	Finite lattice representation: counterexamples and undecidability	Colored-graph characterization: a finite lattice is the congruence lattice of a finite algebra iff it admits a finite colored-graph witness.	No non-representable lattice is exhibited; undecidability and subgroup-interval conclusions explicitly outside. Only the supporting characterization is formalized.
207	The ℓ^1-Bass and complex Bass trace conjectures	Complex Bass trace conjecture for $C[G]$, every group; for torsion-free G , trace equals augmentation rank and char-0 idempotent conjecture.	The $\ell_1(G)$ Bass conjecture (headline, first in title) is not formalized; only the weaker complex group-ring version is.
210	Foulkes' conjecture for sixth powers and quadratic stabilization	Canonical Foulkes-Howe map surjective for $a \geq 2$, $b \geq a(a-1)$, all finite-dimensional V ; hence sixth-power embedding only for $b \geq 30$.	Sixth-power Foulkes conjecture for $6 \leq b < 30$ (companion paper's range; the title's first claim) is not included.
211	Geometry, diffusion, and spectra of random planar maps	Only $q > 4$: rescaled critical finite FK maps converge in Gromov-Hausdorff-Prokhorov distribution to the Brownian continuum random tree.	LQG convergence for $0 < q \leq 4$, the surface-to-tree transition, Liouville Brownian motion and eigenvalue/heat-trace convergence are unformalized.
212	No bigeodesics and smooth limit shapes in planar first-passage percolation	Differentiability of the time-constant norm (C^1 unit sphere) for every Gamma edge-weight law, including exponential with unique supporting lines.	Nonexistence of bigeodesics (title's headline) is not formalized; strict convexity of the exponential limit shape explicitly outside.

#	Title	What Lean proves	Not formalised
215	Massive continuum limits and exact mass asymptotics for planar $O(n)$ models	Finite-volume exponential decay of spin correlations for 2D classical $O(n)$, $n \geq 3$, uniform over finite subgraphs with bounded couplings.	$O(3)$ continuum limit/mass gap and $O(4)$ exact mass asymptotics (the title claims) not formalized; infinite-volume limit also excluded.
218	Conformal universality for weakly interacting and disordered Ising models	Finite-graph lemma: Ising conditional likelihood ratios with pinned spins bounded by $\exp(+4 \operatorname{artanh} q_0)$, q_0 an FK connection probability.	All universality results (spin, energy, SLE_3 interface limits, disordered case) are unformalized; only a supporting comparison inequality is.
222	Perceptron free energies and microscopic jamming	Spherical perceptron pressure formula for bounded continuous potentials; Ising perceptron only finiteness of the variational value; supporting dual formula.	Ising perceptron free-energy convergence and Borel potentials, the bi-orthogonally invariant extension, and jamming (in title) are not formalized.
227	The dynamical phase transition in the Sherrington–Kirkpatrick model	Spectral gap for $\beta < 1$; ratio cutoff only for $\beta < 1/2$; critical mixing between $n^{2/3-\epsilon}$ and $e^{\epsilon n}$; $\beta > 1$ stretched-exponential barrier.	Cutoff for $1/2 \leq \beta < 1$ and its location; critical $n^{2/3+o(1)}$ upper bound (only $e^{\epsilon n}$ proved); critical autocorrelation limits.
229	Sharp three- and four-state reconstruction thresholds	Only the supercritical direction for the symmetric 3-state channel: reconstruction when $b \cdot \lambda^2 > 1$ (regular) or $d \cdot \lambda^2 > 1$ (Poisson trees).	Non-reconstruction at/below the threshold (the sharpness), the four-state case and stochastic-block-model consequences are not formalized.
230	Exact Hausdorff measure for SLE	Positivity only: every gauge agreeing with $r^d (\log \log 1/r)^{(2-d)/2}$ near 0 gives a.s. positive measure to every positive-time segment.	Finiteness of the gauge measure and finite expected measure in bounded disks are not formalized; exactness requires both bounds.
237	The three-quarter diameter exponent for honeycomb walks	Supporting results: finite critical bridge masses, existence of the free-energy limit, strip-crossing mass $\sim N^{-1/4}$ and displacement moment $\sim N^{3/4}$.	The 3/4 diameter exponent and 4/3 mass/covering exponents (the headline) are not formalized; only ingredients from 3 of 13 papers.
240	Shelah's eventual categoricity conjecture	Only the CH obstruction: an abstract elementary class that is not categorical at one prescribed threshold cardinal but is categorical above a larger one.	The eventual categoricity theorem itself (the title's headline) is not formalized.
247	An infinite finitely presented residually finite 2-group	An infinite finitely presented periodic group (a Steinberg group over an F2-algebra), negatively answering the finitely presented Burnside question.	Residual finiteness and 2-power element orders (both in the title) are not asserted; nil-algebra and Kurosh conclusions outside.
256	Howie's conjecture on equations over groups	Kervaire-type coefficient injectivity: for a relator in A^*Z with exponent sum $\neq \pm 1$, A embeds in the one-relator quotient.	Howie's conjecture for systems with full-rank exponent matrix (the title headline) is not formalized; only the one-relator Kervaire companion.
260	Spacetime Penrose inequalities and rigidity	Area-controlled replacement of 3D CKS hyperboloidal ends by asymptotically flat ends, ADM masses approaching Bondi mass; Schwarzschild equality examples.	No Penrose inequality is formalized (main inequality explicitly unformalized; only a conditional connected case retained); rigidity and charged bounds missing.
261	Localization and delocalization in the Anderson model	Supporting fact: the 2D Anderson operator is a.s. bounded self-adjoint with spectrum equal to $[-4-h, 4+h]$.	Pure-point spectral type in 2D and absolutely continuous spectrum in $d \geq 3$ (both headline claims) are not formalized.
263	The ionization and generalized ionization conjectures	Thomas-Fermi asymptotics $I_m(Z)/m^{7/3} \rightarrow a_{\text{TF}}$ as $m, Z/m \rightarrow \infty$; iterated asymptotics of outer-electron radii of neutral atoms.	The ionization conjecture (Z +CM bound) and universal bounds on ionization energies and radii are not formalized; only generalized asymptotics.
266	Exactly three mutually unbiased bases in dimension six	Weaker bound: at most five MUBs in $\mathbb{C}^6$; Fourier character-sum vanishing for non-Tao order-6 Hadamards; cube-fiber cancellation lemma.	The bound $N(6) \leq 3$ (exclusion of four bases), the headline, is explicitly not established.

#	Title	What Lean proves	Not formalised
267	Positive-temperature Bose–Einstein condensation and quantum depletion	Ground-state (zero-temperature) Bose-Einstein condensation in the dilute hard-sphere gas: condensate fraction $\geq c_0$ when $\rho^* a^3 < \varepsilon_0$.	Positive-temperature condensation (the headline) and the Bogoliubov leading quantum-depletion law are not formalized.
271	Bloch's law and spontaneous ferromagnetic order	Spontaneous magnetization: nearest-neighbor Heisenberg ferromagnet on $\mathbb{Z}^d$, $d \geq 3$, any spin, has low-temperature KMS state with magnetization $\geq S/4$.	Bloch's $T^{3/2}$ law with exact coefficient (the headline) and the first lattice correction are not formalized.
272	Entanglement without secret key and the PPT-square conjecture	PPT-square counterexample: PPT channel on 21×21 matrices whose square is not entanglement breaking; dimension-ten PPT pair composition.	The zero-distillable-secret-key entangled state (the headline, first in title) is explicitly outside the formalization.
281	QAOA optimality for the SK model	Parisi-value identities and finite Gaussian approximation, conditional on an admissible Parisi minimizer; full support of zero-temperature minimizers.	Convergence of QAOA circuit energies is not asserted; results are supporting ingredients and conditional on minimizer existence; MaxCut corollary absent.
290	Connes' bicentralizer conjecture and relative bicentralizers	A bounded-recovery lemma converting positive modular spectral averages into uniformly bounded algebra elements along a subsequence.	The absolute and relative bicentralizer conjectures themselves are not formalized; only a supporting lemma.
291	Toms–Winter and equivariant Jiang–Su stability	Challenge: tracial real rank zero of the uniform tracial ultrapower implies uniform property Gamma; linked results give Z-absorption under strict comparison.	Equivariant Jiang-Su stability (Szabo's conjecture case) is not formalized; full Toms-Winter equivalence with nuclear dimension not a selected statement.
307	Failure of rational injectivity for maximal coarse assembly	Graph-union space whose ordinary (reduced Roe algebra) coarse assembly map kills an infinite-order class that survives rationally.	Maximal assembly failure (title claim, strictly stronger) not formalized; only the companion reduced/coarse Novikov counterexample.
312	The Grothendieck homotopy hypothesis	Elementary-expansion theorem: attaching an $(n+1)$ -disk along its source n -face is a weak equivalence for boundary-cellular models.	Semi-model structure and the comparison with homotopy theory of spaces (the hypothesis itself) are not formalized.
323	Relative independence of the separable quotient problem	Under CH, real and complex infinite-dimensional Banach spaces with no infinite-dimensional separable quotient.	Positive consistency direction (from a measurable cardinal) and hence the independence result are not formalized.
337	Cartan–Hadamard isoperimetry and CAT(0) fillings	Sharp Euclidean filling inequality for compactly supported integral n -cycles ($n \geq 2$) in proper CAT(0) spaces, and optimality of the constant.	Manifold isoperimetry for finite-perimeter sets, the $\kappa < 0$ model comparison, and the equality case are not formalized; only a $\kappa = 0$ integral-current form.
347	Counterexamples to strong forms of Arnold's fixed-point conjecture	Hamiltonian diffeomorphism of the quadric threefold with exactly three fixed points (one degenerate) versus four critical points.	The headline nondegenerate stable-Morse counterexamples on 22-dimensional Kahler manifolds with unbounded deficit are not formalized.
348	Nonnegatively curved Einstein four-manifolds and a topological gap	Classification only under strictly positive sectional curvature: round S^4 , Fubini-Study $\mathbb{C}P^2$, or $\mathbb{R}P^4$.	Nonnegative-curvature case (including $S^2 \times S^2$) and the trace-free Ricci L^2 topological gap theorem are not formalized.
356	Gigli's characterization of Alexandrov curvature	Transfer of a weak Hessian upper bound to every constant-speed geodesic in $\text{RCD}(K, N)$ spaces.	The characterization of Alexandrov curvature itself is not formalized; only a supporting weak Hessian lemma.
363	Boltzmann nonuniqueness with exact local conservation	Two distinct renormalized solutions with same data, local mass and total momentum conservation, energy/entropy inequalities, L^1 continuity initially.	Exact local momentum and energy conservation (the title's distinctive claim) and global L^1 continuity are not formalized.

#	Title	What Lean proves	Not formalised
365	Joint metric and connection recovery from one boundary patch	Two distinct bounded measurable scalar conductivities on a 3D ball with identical full Dirichlet-to-Neumann map.	The headline joint metric-and-connection recovery theorem is not formalized; only the contrasting conductivity nonuniqueness result.
367	The critical dimension for the one-phase Bernoulli problem	Existence of a nonflat one-homogeneous global minimizer in dimension seven.	Flatness of minimizers in dimensions ≤ 6 (the new part) and resulting regularity and singular-set bounds are not formalized.

Minor gaps: headline formalised, part of the claim not

#	Title	Not formalised
039	Nagata's conjecture and maximal Seshadri constants	Maximal Seshadri constants in dimension $n \geq 3$ and in positive characteristic are not formalized; only complex surfaces.
047	Complex counterexamples to cancellation and affine fibrations	The Dolgachev-Weisfeiler affine-fibration counterexample (non-locally-trivial A^3 -fibrations), named in the title, is not formalized.
082	Maximal and variational bounds for the triangular Hilbert transform	The r -variational bounds named in the title, a.e./norm convergence, and the explicit trilinear $L^3 \times L^3 \times L^3$ estimate are not stated.
088	Petty's projection-volume conjecture and simplex counterexamples	The full Lutwak-Petty projection inequalities and exponential-factor counterexamples in every sufficiently large dimension are not formalized.
090	Universal optimality of the triangular lattice	Renormalized Riesz ($0 < s < 2$) and Coulomb minimality, hence the Sandier-Serfaty and Brauchart-Hardin-Saff conjectures, are not formalized.
091	The logarithmic Brunn–Minkowski conjecture	The B-conjecture for even log-concave measures and the L_p Brunn-Minkowski inequality for $0 < p < 1$ are not stated.
107	Matrix multiplication with exponent at most $9/4$	The $\omega < 2.258$ bound for all but finitely many positive characteristics is not formalized.
116	Uniform identity testing for noncommutative formulas	Positive characteristics are not covered, and the polynomial-time construction bound is not asserted for the polynomial-formula tuple.
119	The Courtade–Kumar and Hellinger conjectures	Hellinger conjecture (square-root loss bound, separate paper) is named in the title but not formalized.
122	Superpolynomial lower bounds and quasipolynomial reconstruction from deletion traces	Quasipolynomial-time/sample reconstruction algorithm (named in title) and polynomial small-deletion regime are not formalized.
131	Polynomial mixing of graph switches with prescribed degrees	Exactly uniform sampling algorithm with expected polynomial bit time is explicitly outside scope.
161	Counterexamples to Sidorenko's conjecture and the forcing conjecture	Forcing-conjecture disproof (named in title) is explicitly outside the statement.
162	Counterexamples to Ryser's covering and Gyárfás's tree-cover conjectures	Gyarfas monochromatic tree-cover conjecture disproof (named in title) is not stated in Lean.
175	Talagrand's conjectures and graph decompositions at expectation thresholds	Graph-decomposition application (named in title, third paper) is not formalized.
183	Power savings for planar halving lines and k-sets	The k -dependent k -set bound $n(k+1)^{1/3-\epsilon_0}$ (title's k -sets) is not formalized; only a uniform $n^{4/3-\epsilon}$ bound with extra genericity.
186	The Friedgut–Kalai graph and hypergraph threshold conjectures	Hypergraph case (named in title, other paper) and nonmonotone influence bound not formalized.
199	Counterexamples to conjectures of Auslander–Reiten, Tachikawa, and Nakayama	Endomorphism-algebra counterexamples to the Nakayama conjectures (named in title), Auslander-Gorenstein and Wakamatsu; Tachikawa field-extension persistence.
220	Directional zero–one laws and ballisticity in random environments	The zero-one law for stationary ergodic finite-range-dependent uniformly elliptic environments (unformalized third paper) is missing.
235	Random-SAT thresholds, sharp variance and computability	For $k=3$ the hitting-time variance is only shown between $\Omega(n)$ and $O(n \log n)$, not the claimed $\Theta(n)$.
244	The partition principle does not imply choice	The ctm construction needs an extra inaccessible cardinal; 'no new countable sequences' preservation is outside.
251	Amenability, unitarizability, and Ulam stability	The strong Ulam stability characterization of amenability (named in the title; second paper) is not formalized.
262	Sharp one-dimensional Lieb–Thirring inequalities	Matrix-valued potentials (constant independent of matrix size) and the full characterization of all equality cases are not formalized.
269	The Laughlin gap and stability under scalar disorder	Stability of the gap and uniqueness of the ground state under weak projected scalar potentials (title's second claim) not formalized.
276	The classical capacity of generalized amplitude damping	Additivity when tensoring with an arbitrary different partner channel (named catalogue claim) is not formalized.

#	Title	Not formalised
280	Strong locality for strongly rational unitary vertex operator algebras	Complete rationality of the net, unitarizability of simple modules, and the braided unitary tensor equivalence are not formalized.
299	The Kirchberg–Rørdam character criterion	Dadarlat-Toms consequence (infinite tensor power of character-free algebra is Z-stable) is not formalized.
350	Yau's nodal conjecture: surfaces and counterexamples	Dimension-five power-law growth faster than $\lambda^{1/2+\epsilon_0}$ not formalized; only an unbounded nodal-measure/ $\sqrt{\lambda}$ ratio.
353	The sharp dimension threshold for affine Bernstein rigidity	Sharpness (dimension-10 nonquadratic example, companion paper) and the affine-metric-complete hypersurface extension are not formalized.
361	Counterexamples to Yau's harmonic dimension bound	The $\mathbb{R}^3$ construction valid for every sufficiently large k with $(k+2)^2$ functions is not formalized.
373	Nonattainment in three-marginal Coulomb transport	Extension to every inverse-power Riesz exponent in every dimension ≥ 2 is not formalized.

7 Key insights

1. Trust has to be judged result by result

The repository mixes machine-checked theorems, theorems where Lean checks something weaker than the title, and unrefereed papers, often within one family. For the headline claims discussed in this guide:

☒ LEAN-CHECKED	☐ PARTLY LEAN	☐ PAPER ONLY
ζ and Dirichlet L zero-free for $\text{Re } s > 7/8$ (003)	Fourier transform below $n \log n$ (130)	BSD in ranks 0 and 1 (002)
Irrationality exponent of π (017)	Exactly 3 MUBs in dimension 6 (266)	Hodge for CM abelian varieties (032)
Catalan's constant irrational (005)	Anderson localisation (261)	Modularity over imag. quadratic fields (030)
Unique Games Conjecture (102)	BEC at positive temperature (267)	Hilbert's tenth problem over $\mathbb{Q}$ (004)
$\omega \leq 9/4$ (107)	Ionization conjecture (263)	$L = RL = BPL$ (103)
Mahler conjectures (087)	Quasipolynomial Szemerédi bounds (159)	Subset Sum below $2^{n/2}$ (138)
Thompson's group F (248)		Hilbert–Smith conjecture (304)
Free group factors (287)		Haldane gap (268)
Kadison similarity (288)		BFSS bound states (270)
Cannon's conjecture (246)		Strong cosmic censorship near Kerr (264)
Plane not 5-colourable (158)		
Borsuk fails in dimension 9 (156)		
Polyakov decay ($O(n)$, $n \geq 3$) (215)		
Critical percolation on $\mathbb{Z}^3$ (213)		
Laughlin gap (269)		
Heisenberg ferromagnet order (271)		
Relativistic Vlasov–Maxwell (362)		

Read the first column as “true, provided the formal statement says what the paper says”. The second column is the easiest to misread: a Lean proof exists, but of a weaker or supporting statement. The third column holds serious but unrefereed preprints.

2. A formal proof turns refereeing into statement-checking

Nobody can referee 25.9 million lines, but anyone with some Lean experience can read a one-line zeta statement or a three-line colouring statement and decide whether it matches the paper. OpenAI's scope notes in `lean/docs` make this easier still, because they say plainly what each formalisation leaves out. The remaining risks are definitions that subtly differ from the paper's, statements weaker than the title (Section 6), and the fact that OpenAI marks its formalisations as “unchecked”.

3. Formal coverage is thinnest in algebraic geometry and topology

The share of families with Lean ranges from 100% in mathematical logic, 91% in functional analysis and 89% in combinatorics, down to 52% in number theory, 19% in algebraic geometry and 17% in topology. The likely reason is Lean's library, Mathlib, which has deep foundations for analysis, combinatorics and algebra but little of the machinery modern arithmetic and algebraic geometry rely on (étale cohomology, automorphic forms, Hodge theory). That is why BSD, Hodge and modularity have no formal proof, while analytic number theory (zeta, π , Catalan) does.

4. Disproofs are easier to certify than proofs

About 64 families are counterexamples or other negative results; 47 of them (73%) have Lean pages, against 188 of 308 (61%) positive results (a keyword-based estimate). A counterexample is a concrete object a computer can check; a positive theorem usually needs a whole supporting theory.

5. The model builds on its own results, so errors can be correlated

Results cite each other. The π reasoning summary draws on “OpenAI's Catalan determinants” (family 005); BSD (002) combines with Goldfeld (006); Milne's rationality conjecture (001) uses the Hodge result (032). This compounding is powerful, but a flaw in one unformalised base result would spread to everything built on it. Here the Catalan and π results are themselves Lean-checked, but BSD, Goldfeld and Hodge are not.

6. Inside a reasoning summary: how the π result was found

The π summary runs to 42 pages and 38 sections, most of them dead ends: Padé approximants, modular forms, elliptic periods, BBP formulas, theta functions and more, each abandoned when a height or denominator cost could not be paid. The model was initially aiming only for $\mu(\pi) < 5/2$ (enough for Flint–Hills) and found a candidate bound of $62/25 = 2.48$. Then, while “stress-testing ... its unexpectedly strong exponent”, it noticed its method could be pushed down to 2. The final result is now Lean-checked. These are abridged summaries, not full transcripts.

7. The flagship results came from a different, undisclosed process

The README excludes the zeta zero-free region and the Hodge result for CM abelian varieties from its standard “about 3 hours per result” procedure, and says the 11/12 write-up was edited by humans. How those were produced is not stated. Of the two, the zeta result is fully Lean-checked; the Hodge result has no formal proof.

8. OpenAI's own top-level catalogue understates its formal coverage

lean/formalization.yaml lists 178 of the 405 Comparator challenges and 162 of the 330 formalised papers. The first edition of this guide trusted it and reported Lean proofs for only 127 families. The per-family pages in lean/docs are complete. Anyone summarising this release should start there.

9. Production has outrun review

With 370 manuscripts dated within two days, normal refereeing cannot keep up. Signs of a field adjusting in real time include readability complaints about the earlier Navier–Stokes paper despite its Lean check, credit disputes, Tao's advisory group and the Hexagon repository launched on release day, five family numbers that silently disappeared, and the October 2025 episode in which claimed “solutions” turned out to be literature lookups. Expect corrections; OpenAI says it will publish them as new versions.

A practical checklist for any claim in this repository

- Find the family in Appendix A: it shows how many papers are formalised and flags any gap with the title.

- If it has Lean: open [lean/docs/NNN.md](#) and read the scope note, then read the linked challenge statement in [lean/ComparatorChallenges/](#) and compare it with the paper's abstract. Note any custom definitions.
- If not: treat it as an unrefereed preprint. Check whether its catalogue entry depends on other families ("Together with result ..."), and how those are verified.
- Look for independent follow-up: expert commentary, reuse in other papers, Hexagon, or arXiv audits like the one done for the August results.

8 What this review did, and what changed in this edition

Did: confirmed provenance on GitHub; parsed all 372 catalogue entries; read OpenAI's Lean page for all 235 formalised families; checked all 405 Comparator configurations; scanned every Lean file for proof-skipping constructs; mapped every challenge to its family twice (from OpenAI's pages and by an independent statement-by-statement review, which agreed completely); read about forty challenge statements in full, including every headline in Sections 4 and 5; audited every formalised family for gaps against its title (four independent reviewers, spot-checked against the statements I read); and read the π reasoning summary.

Did not: compile the Lean library; check any paper's mathematics; or read most of the 722 manuscripts. Background facts about earlier work (previous records and dates) are from my own knowledge and worth double-checking. Outside reactions come from search-engine summaries, because several sites were blocked from my environment. The gap verdicts are judgements made from OpenAI's scope notes; borderline cases were resolved conservatively (towards "gap").

Corrections to the first edition (6 October 2026)

- **Lean coverage.** Reported as 127 families and 162 papers; actually 235 families and 330 papers. The first edition relied on `lean/formalization.yaml`, which omits 227 of the 405 challenges.
- **Badges that changed from "paper only" to Lean-checked:** the irrationality exponent of π (017), Catalan's constant (005), the free group factors (287), Kadison's similarity problem (288), Cannon's conjecture (246), the general Mahler conjecture (087), and the Dirichlet and Hecke versions of the $7/8$ zero-free region (003).
- **Badge that changed to "partly Lean":** the Fourier transform below $n \log n$ (130), whose Lean statement holds only along a sequence of lengths.
- **Physics.** The first edition implied physics was the least verified area; in fact 17 of 25 mathematical-physics families have Lean pages. The least-covered areas are algebraic geometry and topology.
- **New in this edition:** the physics section (Section 5), the gap audit (Section 6), the "most formally verified" table (Section 2), the re-run code scan (Section 3), and gap flags and Lean challenge names in Appendix A.

Appendix A All 372 result families

Every family in OpenAI's catalogue, grouped by subject in catalogue order. Titles link to the first paper of each family on GitHub; the grey line beneath lists the family's Lean Comparator challenges (files in `lean/ComparatorChallenges/`). "Lean" shows how many of the family's papers are formalised according to OpenAI's `lean/docs` pages, and flags gaps found by the audit in Section 6. A "guide" link jumps to that family's card.

Number theory · 31 families, 16 with Lean

#	Result (and its Lean challenges)	Papers	Lean
001	Milne's rationality conjecture	1	—
002	The full Birch–Swinnerton-Dyer formula in Selmer coranks zero and one ▶ guide	3	—
003	The quasi-Riemann hypothesis ▶ guide <code>QuasiRiemannHypothesis</code> , <code>DirichletSevenEighths</code> , <code>HeckeSevenEighths</code> , <code>SiegelZeros</code>	3	● 2 of 3
004	Hilbert's tenth problem over $\mathbb{Q}$ ▶ guide	2	—
005	Irrationality of Catalan's constant ▶ guide <code>Catalan</code>	1	● 1 of 1
006	Goldfeld's conjecture ▶ guide	2	—
007	Two-point Chowla and the corrected binary Elliott conjecture <code>OrdinaryElliott</code> , <code>OrdinaryTwoPointCorrelations</code>	1	● 1 of 1
008	The Deligne–Drinfeld conjecture <code>DeligneDrinfeld</code>	1	● 1 of 1
009	Bogomolov–Pop and Milnor K-theoretic reconstruction <code>BogomolovPopInjectivity</code>	3	● 1 of 3 major gap
010	Fontaine–Mazur modularity at the prime 2 and 2-adic pro-modularity	3	—
011	The Ford–Konyagin–Luca conjecture on prime predecessors	3	—
012	Independent largest prime factors of consecutive integers <code>JointDickman</code>	1	● 1 of 1
013	Ostmann's inverse Goldbach conjecture <code>OstmannComplete</code> , <code>OstmannPrimes</code>	1	● 1 of 1
014	Restricted geometric Langlands, generic Ramanujan and Arthur parameters	8	—
015	Torus-packet equidistribution in prime, quartic, and sextic degrees <code>DukePrimeDegree</code>	3	● 1 of 3 major gap
016	Zilber–Pink in abelian varieties and the Siegel threefold	4	—
017	The irrationality exponent of π is 2 ▶ guide <code>PiExponent</code>	1	● 1 of 1
018	The Margulis–Platonov conjecture over global fields	2	—
019	The p-adic section conjecture	2	—
020	Squarefree quartics and power-free polynomial values <code>PowerFreeValues</code>	1	● 1 of 1
021	A quadratic bound for Jacobsthal's function <code>Jacobsthal</code> , <code>JacobsthalImproved</code>	1	● 1 of 1
022	The weak inhomogeneous Duffin–Schaeffer conjecture	1	—
023	Patterson's first moment for cubic Gauss sums <code>PattersonFirstMoment</code>	1	● 1 of 1
024	An asymptotic formula for the number of totients <code>TotientAsymptotic</code> , <code>TotientCompanionZero</code>	1	● 1 of 1
025	Erdős's short Egyptian-fraction conjecture <code>EgyptianFractions</code> , <code>ShortEgyptianFractions</code>	1	● 1 of 1
026	Positive lower density of large prime gaps <code>PrimeGaps</code>	1	● 1 of 1
027	Integral density on curve character varieties	1	—
028	The Gaussian moat conjecture ▶ guide <code>GaussianMoat</code>	1	● 1 of 1
029	Artin's primitive root conjecture: infinitude for every base ▶ guide	2	—
030	Modularity over imaginary quadratic fields ▶ guide	1	—
031	Uchida's conjecture for open Galois homomorphisms	1	—

Algebraic and complex geometry · 36 families, 7 with Lean

#	Result (and its Lean challenges)	Papers	Lean
032	The rational Hodge conjecture for CM abelian varieties and products of K3 surfaces ▶ guide	8	—
033	Campana's orbifold litaka conjecture and logarithmic subadditivity <code>LogKodairaFiberNegative</code>	5	● 1 of 5 major gap
034	Log abundance and effective litaka fibrations	14	—
035	Threefold log abundance in numerical dimension one in characteristic $p > 3$	2	—
036	Numerical semiampleness and generalized minimal models	4	—
037	The sharp ordinary-double-point volume gap	2	—
038	Fujita's freeness conjecture	1	—
039	Nagata's conjecture and maximal Seshadri constants <code>Nagata</code> , <code>MaximalSeshadriConstants</code>	4	● 2 of 4 minor gap
040	Bloch's conjecture for complex surfaces	1	—
041	Hyperkähler SYZ and projective-space bases	2	—
042	Oka classification for K3 surfaces and other compact complex surfaces	1	—
043	P = W for fixed-determinant moduli spaces	1	—
044	The equivariant cohomological Hikita conjecture for quivers	1	—
046	Counterexamples to Shafarevich holomorphic convexity	2	—
047	Complex counterexamples to cancellation and affine fibrations <code>ComplexCancellation</code>	1	● 1 of 1 minor gap
048	A characteristic-zero counterexample to Lipman–Zariski	1	—
049	A stable-coordinate counterexample in four variables <code>AbhyankarSathaye</code> , <code>CommutingDerivations</code>	2	● 1 of 2 major gap
050	A counterexample to Griffiths' positivity conjecture <code>QuadricBundles</code>	1	● 1 of 1
051	Kobayashi's canonical-ampleness conjecture	1	—
052	Tangent-bundle splittings and universal covers <code>KahlerSplitting</code> , <code>SplitTangentIntegrability</code>	2	● 2 of 2
053	A counterexample to Pixton's original completeness conjecture	1	—
054	Counterexamples to Kuznetsov's rationality conjecture	1	—
055	Toda's Gepner conjecture and large-volume stability	2	—
056	Termination of fourfold minimal model programs	5	—
057	Campana's abelianity conjecture and special varieties	3	—
058	The Kollár–Pardon universal-cover conjecture <code>SymmetricDomains</code>	2	● 1 of 2 major gap
059	Counterexamples to Zariski's multiplicity conjecture	2	—
060	The Global Spherical Shell conjecture	1	—
062	The LeBrun–Salamon conjecture and projective contact classification	1	—
063	The generalized Mukai conjecture	1	—
064	The μ-constant problem for surface singularities	1	—
065	Virasoro constraints for complete intersections and projective bundles	2	—
066	Bounded klt complements for Fano contractions	2	—
067	The Campana–Peternell conjecture in dimension six	1	—
068	Anticanonical nonvanishing under smooth semipositivity	7	—
069	Quantum geometric Langlands at irrational level	1	—

Real and complex analysis · 16 families, 9 with Lean

#	Result (and its Lean challenges)	Papers	Lean
071	Koebe's circle-domain conjecture and circle-domain rigidity KoebeCircleDomains, KoebeCircleDomains	2	● 2 of 2
072	Brennan's conjecture and a counterexample to Kraetzer's prediction Brennan, BrennanSharp, StrictMeans	2	● 2 of 2
073	The Falconer distance conjecture PlanarFalconer, FalconerAllDimensions	1	● 1 of 1
074	Takeya in three and four dimensions	2	—
075	The L log L Fourier-convergence conjecture	1	—
076	Ultraflat real Littlewood polynomials AsymptoticallyMinimalLittlewood, LittlewoodFiniteFlatness	3	● 1 of 3 major gap
077	Fourier restriction for positively curved surfaces	2	—
078	The three-dimensional Bochner–Riesz conjecture	1	—
079	Sogge's local smoothing conjecture in dimension three	1	—
080	The Sobolev endpoint in Carleson's Schrödinger convergence problem	2	—
081	The David–Semmes Riesz-transform problem in higher codimension RieszQuantitative	1	● 1 of 1
082	Maximal and variational bounds for the triangular Hilbert transform TriangularHilbert	3	● 1 of 3 minor gap
083	Stein's conjecture for Hilbert transforms along Lipschitz directions LipschitzHilbert	1	● 1 of 1
084	The geometric case of the Erdős similarity conjecture DyadicAvoidance	2	● 1 of 2 major gap
085	Endpoint regularity of the planar centered maximal function SignedFiniteBand, DiskMaximal	1	● 1 of 1
086	An L^3 bound for the trilinear Hilbert transform	1	—

Convex and metric geometry · 15 families, 13 with Lean

#	Result (and its Lean challenges)	Papers	Lean
087	The Mahler conjectures and symplectic width ► guide MahlerConjecture, SymmetricMahlerEquality, GeneralMahler, SymmetricPolar	3	● 3 of 3
088	Petty's projection-volume conjecture and simplex counterexamples PettyProjectionVolume, ProjectionCounterexample, ProjectionVolume	2	● 2 of 2 minor gap
089	Bounded-distortion L_1 embeddings of planar and bounded-treewidth graphs PlanarL1, BoundedTreewidthL1	2	● 2 of 2
090	Universal optimality of the triangular lattice AtomicGaussian, TriangularEnergy, TriangularGaussian, PlanarPacking	4	● 2 of 4 minor gap
091	The logarithmic Brunn–Minkowski conjecture LogBrunnMinkowski	1	● 1 of 1 minor gap
092	The optimal order of convex-body covering density SingleLatticeCovering, CoveringDensity	2	● 2 of 2
093	Dimension-free logarithmic Sobolev inequality for subgaussian log-concave measures	1	—
094	Subpolynomial dimension reduction in L_p SubpolynomialLp	1	● 1 of 1
095	Hyperbolicity cones without semidefinite lifts HyperbolicCones	3	● 1 of 3 major gap
096	The Gaussian propeller conjecture GaussianPropeller	1	● 1 of 1
097	The Euclidean Steinitz–Bergström conjecture SteinitzBergstrom	1	● 1 of 1
098	A negative answer to the Lang–Plaut problem DoublingHilbert, CompactBanach	1	● 1 of 1
099	The sharp distortion of edit distance into ℓ_1 EditDistance, FiniteCircle, BinaryEditLower, TreeEdit	3	● 3 of 3
100	A counterexample to Bang's cylinder-covering bound CylinderCovering, TriangularCovering, CylinderCovering, RuledCovering, CylinderCovering, CylinderCovering	4	● 4 of 4
101	The sharp simplex conjecture for isotropic constants	1	—

Theoretical computer science · 40 families, 32 with Lean

#	Result (and its Lean challenges)	Papers	Lean
102	The Unique Games Conjecture and optimal approximation thresholds ▶ guide UniqueGamesTheorem, OptimalMaxCut, VertexCover, MinUncut, DirectedFeedback	5	● 5 of 5
103	Derandomization of logarithmic space: $L = RL = BPL$ ▶ guide	1	—
104	Quasipolynomial algorithms for mean-payoff, stochastic and parity games TruffetCounterexample, RandomizedMeanPayoff	4	● 2 of 4 major gap
105	The 2-to-1 Games Conjecture with perfect completeness PerfectCompleteness	1	● 1 of 1
106	Hardness of coloring three-colorable graphs IndependentSets	1	● 1 of 1
107	Matrix multiplication with exponent at most $9/4$ ▶ guide MatrixMultiplication, MatrixFields	3	● 2 of 3 minor gap
108	A cubic permanent–determinant lower bound PermanentCubic, SmoothInitialForm	1	● 1 of 1
109	Integer multiplication below $n \log n$ ▶ guide	1	—
110	Optimal-order randomized k-server on arbitrary metrics KServer, UniformKServer	2	● 2 of 2
111	One-sample matroid prophet inequalities against an almighty adversary MatroidProphet, MatroidSecretary	1	● 1 of 1
112	Beyond the square-root exponent for depth-three circuits DepthThree	1	● 1 of 1
113	Approximate counting and the perfect-matching entropy conjecture MatchingFPRAS, MatchingEntropy, BinaryMatching, SingletonLoopMatching, MatchingEntropyBounds, TriangleFace	2	● 2 of 2
114	Approximate counting of common integer polymatroid bases CommonBasesFPRAS	2	● 1 of 2 major gap
115	Sampling and counting contingency tables with arbitrary margins ContingencyTables, ContingencyTables	2	● 2 of 2
116	Uniform identity testing for noncommutative formulas FormulaHitting, RationalHitting	3	● 2 of 3 minor gap
117	Uniform sparsest cut: hardness and semidefinite gaps UniformSparsestCut	2	● 1 of 2 major gap
118	Unbounded bin-packing gaps and the modified integer round-up conjecture BinPackingGap	1	● 1 of 1
119	The Courtade–Kumar and Hellinger conjectures CourtadeKumar, SoftChannel204	2	● 1 of 2 minor gap
120	Almost-linear-time exact matching in general graphs	1	—
121	Almost-linear expected-time approximation of edit distance EditApproximation	1	● 1 of 1
122	Superpolynomial lower bounds and quasipolynomial reconstruction from deletion traces TraceReconstruction	3	● 1 of 3 minor gap
124	Polynomial-time scheduling on three identical machines ThreeMachine	1	● 1 of 1
125	The approximation threshold for metric k-median KMedianRecovery, KMedianRefinedRecovery, KMedianThreshold, MetricKMedian	2	● 2 of 2
126	Exponential semidefinite complexity of perfect matching MatchingAffineLift, MatchingPSD	1	● 1 of 1 major gap
127	The asymptotic Gotsman–Linial conjecture GotsmanLinial	1	● 1 of 1
128	A factor-two approximation for shortest common superstring Superstring	1	● 1 of 1

#	Result (and its Lean challenges)	Papers	Lean
129	Exponential state costs for two-way automata TwoWayComplementation, TwoWayDeterminization, OneWayLiveness	2	● 2 of 2
130	Fourier transforms below $n \log n$ ► guide ExactFourier	2	● 1 of 2 major gap
131	Polynomial mixing of graph switches with prescribed degrees SwitchChain	1	● 1 of 1 minor gap
132	A counterexample to the quadratic sensitivity conjecture SensitivitySeparation	1	● 1 of 1
133	The complexity of Weisfeiler–Leman refinement ParityLifts, WLIdentification, WeisfeilerLeman, VariableWL	4	● 4 of 4
134	Generalized star height at most three GeneralizedStarHeight, GeneralizedStarHeight	3	● 3 of 3
135	Sharp homogeneous depth-five complexity of matrix products DepthFive	1	● 1 of 1
136	The quasilinear PCP-for-PPAD conjecture	1	—
137	One-tape time simulation in two-fifths-power space	1	—
138	Subset Sum in $O(2^{0.49n})$ time ► guide	2	—
139	Subpolynomial queries for log-concave sampling LogConcaveQuery	1	● 1 of 1
140	Memory–sample lower bounds for noiseless Gaussian regression MemoryPrecision, NoiselessRegression, PosteriorReplicas, GaussianFiniteEntropy, GaussianInformation, ProjectionMoments, GaussianReplacement, SubsphereCurrent	6	● 6 of 6
141	The existential theory of the reals and existential–universal sentences in the counting hierarchy	1	—
142	Deterministic polynomial factorization over prime fields	1	—

Dynamical systems and ergodic theory · 12 families, 9 with Lean

#	Result (and its Lean challenges)	Papers	Lean
143	Uniform limit-cycle bounds in Hilbert's sixteenth problem QuinticLienard	2	● 1 of 2 major gap
144	Banach's simple Lebesgue-spectrum problem ThreeTorus	1	● 1 of 1
145	Rokhlin's multiple-mixing problem Rokhlin	1	● 1 of 1
146	Sinai's positive-entropy conjecture for the standard map StandardMapComponents, StandardMapEntropy, StandardMapLyapunov	1	● 1 of 1
147	The near-boundary Birkhoff conjecture	2	—
148	The dimension formula for self-similar measures on the line SelfSimilar, SelfSimilarCorollaries	1	● 1 of 1
149	Permanence for weakly reversible reaction networks MassAction	2	● 1 of 2 major gap
150	Weak mixing of irrational triangular billiards IrrationalTriangleBilliard	2	● 1 of 2 major gap
151	A C^1 counterexample to Shub's entropy conjecture C1EntropyCounterexample	1	● 1 of 1
152	Zero entropy does not guarantee a smooth positive-volume model SmoothObstruction	1	● 1 of 1 major gap
153	Arithmetic classification of Bernoulli convolutions	1	—
154	Pointwise multiple ergodic averages for mixing transformations	4	—

Combinatorics · 37 families, 33 with Lean

#	Result (and its Lean challenges)	Papers	Lean
155	A counterexample to periodic tiling in dimension three ► guide PeriodicTilingThree	1	● 1 of 1
156	Borsuk's conjecture fails in dimension nine ► guide BorsukNine	1	● 1 of 1
157	Counterexamples to the Hadwiger and Colin de Verdière conjectures ListHadwiger	3	● 1 of 3 major gap
158	The Euclidean plane cannot be colored with five colors ► guide EuclideanFiveColor, PlaneColoring	1	● 1 of 1
159	Erdős's reciprocal-sum conjecture and quasipolynomial Szemerédi bounds ErdosReciprocal	1	● 1 of 1 major gap
160	Superexponential van der Waerden numbers QuantitativeVanDerWaerden	1	● 1 of 1
161	Counterexamples to Sidorenko's conjecture and the forcing conjecture SidorenkoCounterexample	1	● 1 of 1 minor gap
162	Counterexamples to Ryser's covering and Gyárfás's tree-cover conjectures BalancedRyser, RyserCovering, RyserOddExtensions	2	● 2 of 2 minor gap
164	Hindman's finite sums and products conjecture	1	—
165	Exact crossing numbers of complete and complete bipartite graphs CompleteCrossing, BipartiteCrossing	2	● 2 of 2
166	The higher-dimensional Erdős distinct-distances conjecture	1	—
167	Pinned distances and a power saving for planar unit distances PinnedDistances, PlanarUnitDistances	2	● 2 of 2
168	Combinatorial invariance of Kazhdan–Lusztig polynomials KLInvariance	1	● 1 of 1
169	The Shreshian–Wachs e-positivity conjecture ElementaryPositivity	1	● 1 of 1
170	Sharp logarithmic exponents for off-diagonal Ramsey numbers RamseyFive, SharpLogRamsey	2	● 2 of 2
171	The hypercube Ramsey conjecture	1	—
172	Classification of finite Euclidean Ramsey configurations EuclideanRamsey, GrahamSpherical, EuclideanRamseyCircle, EuclideanRamseyNine, EuclideanRamseyQuadratic, EuclideanRamseySpherical, EuclideanRamseyTransitive	1	● 1 of 1
173	Seymour's second-neighborhood conjecture SeymourSecondNeighborhood	1	● 1 of 1
174	Deterministic construction of strong thin spanning trees StrongThinTree, AlgorithmicThinTrees	2	● 2 of 2
175	Talagrand's conjectures and graph decompositions at expectation thresholds TalagrandExpectationThreshold, TalagrandDiscreteConvexity	3	● 2 of 3 minor gap
176	The second Kahn–Kalai conjecture SecondKahnKalai	1	● 1 of 1
177	Bounded-degree coboundary expanders in every dimension CoboundaryExpanders	1	● 1 of 1
178	Deterministic nonbipartite Ramanujan graphs in every fixed degree	1	—
179	The circulant Hadamard conjecture CirculantHadamard, EvenBarker	1	● 1 of 1
180	Barnette's Hamiltonian-cycle conjecture BarnetteHamiltonian	1	● 1 of 1
181	The Erdős–Gallai cycle-decomposition conjecture CycleDecomposition	1	● 1 of 1

#	Result (and its Lean challenges)	Papers	Lean
182	Power savings for polynomial-difference-free sets SquareDifference	3	● 1 of 3 major gap
183	Power savings for planar halving lines and k-sets HalvingLines	1	● 1 of 1 minor gap
184	Coloring and independence in graphs with forbidden subgraphs CliqueFreeLog	2	● 1 of 2 major gap
185	Counterexamples to infinite matroid intersection and packing/covering InfiniteMatroid, InfiniteMatroidCorollaries	1	● 1 of 1
186	The Friedgut–Kalai graph and hypergraph threshold conjectures SharpThreshold	2	● 1 of 2 minor gap
187	Snaky in 21 Maker moves SnakyCertificate, SnakyConditional, SnakyTwentyOne	1	● 1 of 1
188	The sharp constant in random triangle removal TriangleRemoval	1	● 1 of 1
189	Exact cycle–clique Ramsey numbers CycleCliqueRamsey	1	● 1 of 1
190	Polynomial removal fails for ordered binary matrices MatrixRemoval	1	● 1 of 1
191	A power improvement in the Heilbronn triangle problem HeilbronnTriangle	1	● 1 of 1 major gap
192	A counterexample to the Gopalan–Servedio conjecture SquareRootDegree	1	● 1 of 1

Algebra · 18 families, 9 with Lean

#	Result (and its Lean challenges)	Papers	Lean
193	Serre's intersection-multiplicity conjecture	1	—
194	Lech's multiplicity conjecture DuttaDomain	1	● 1 of 1 major gap
195	A counterexample to the small Cohen–Macaulay module conjecture	1	—
196	A counterexample to Kaplansky's zero-divisor conjecture ► guide TorsionFreeZeroDivisors	1	● 1 of 1
197	Nonsofic groups and group-ring counterexamples KaplanskyFinitelyPresented, KaplanskyDirectFiniteness, GroupRingDeterminant, OddKaplansky	4	● 3 of 4 major gap
198	A counterexample to the little finitistic-dimension conjecture LittleFinitistic, FinitisticAsymmetry	1	● 1 of 1
199	Counterexamples to conjectures of Auslander–Reiten, Tachikawa, and Nakayama AuslanderReiten, Tachikawa	2	● 2 of 2 minor gap
200	Eisenbud–Green–Harris and lex-plus-powers in characteristic zero	2	—
201	A counterexample to Kurosh's division-ring problem	1	—
202	The blockwise Alperin weight conjecture	1	—
203	Donovan's conjecture over fields and discrete valuation rings	2	—
204	Tensor saturation for even spin groups	1	—
205	Saxl's conjecture and universal tensor squares UniversalTensorSquares, Saxl	2	● 2 of 2
206	Finite lattice representation: counterexamples and undecidability FiniteCongruenceGraph	2	● 1 of 2 major gap
207	The ℓ^1-Bass and complex Bass trace conjectures BassTrace, BassTorsionFree	2	● 1 of 2 major gap
208	The finite Benson–Etingof–Ostrik conjecture	1	—
209	Integral counterexamples to Gersten's conjecture	2	—
210	Foulkes' conjecture for sixth powers and quadratic stabilization FoulkesHowe	2	● 1 of 2 major gap

Probability and statistical mechanics · 29 families, 19 with Lean

#	Result (and its Lean challenges)	Papers	Lean
211	Geometry, diffusion, and spectra of random planar maps FKCRT	7	● 1 of 7 major gap
212	No bigeodesics and smooth limit shapes in planar first-passage percolation PlanarFirstPassage, GammaPassage	2	● 1 of 2 major gap
213	No infinite critical clusters on quasi-transitive graphs ▶ guide CriticalZ3, CriticalPercolation	2	● 2 of 2
214	The Benjamini–Schramm nonuniqueness conjecture BenjaminiSchramm, CayleyPercolation	1	● 1 of 1
215	Massive continuum limits and exact mass asymptotics for planar $O(n)$ models ▶ guide ClassicalON	5	● 1 of 5 major gap
216	Critical logarithmic corrections and BKT scaling for the planar XY model	6	—
217	The low-temperature Sherrington–Kirkpatrick fluctuation law	2	—
218	Conformal universality for weakly interacting and disordered Ising models BufferedIsing	6	● 1 of 6 major gap
219	GOE universality for random regular graphs with weak disorder	2	—
220	Directional zero–one laws and ballisticity in random environments DirectionalWalk, DirectionalBallisticity, VelocityHemisphere	3	● 2 of 3 minor gap
221	The Mézard–Parisi formula for diluted spin glasses DilutedSpin	1	● 1 of 1
222	Perceptron free energies and microscopic jamming IsingFiniteness, PerceptronFreeEnergy, SphericalField	4	● 2 of 4 major gap
223	Conformal limits of square-lattice random-cluster interfaces	6	—
224	Critical and near-critical universality for Voronoi percolation	3	—
225	Gaussian free field limits for the balanced six-vertex model	1	—
226	The double-dimer CLE_4 conjecture in the half-plane	1	—
227	The dynamical phase transition in the Sherrington–Kirkpatrick model SKHighTemperature, SKRatio, CriticalSKMixing, CriticalSK, SKBarriers	8	● 4 of 8 major gap
228	Continuum phase transitions for radial pair potentials ContinuumTransition, RadialTransition, RadialDensityInterval	2	● 2 of 2
229	Sharp three- and four-state reconstruction thresholds ThreeStateSupercritical, ThreeStateTreeClauses	3	● 1 of 3 major gap
230	Exact Hausdorff measure for SLE SLELowerPositivity	2	● 2 of 2 major gap
231	The free uniform spanning forest is a factor of IID StronglyRayleighDPP, FreeUniformSpanningForest	1	● 1 of 1
232	Gaussian fields and SLE interfaces for Lipschitz heights	3	—
233	The joint critical Ashkin–Teller scaling limit	1	—
234	All-temperature pressure for orthogonally invariant Ising spin glasses InvariantIsing	1	● 1 of 1
235	Random-SAT thresholds, sharp variance and computability FixedClauseThreshold, SATSharpness, SATVariance, SATComputability	4	● 3 of 4 minor gap
236	The factor-of-IID threshold for free Ising states on trees FreeIsing	1	● 1 of 1
237	The three-quarter diameter exponent for honeycomb walks HoneycombBridgeFiniteness, HoneycombFreeEnergy, CriticalStripMass	13	● 3 of 13 major gap
238	Optimal logarithmic mixing of the Thorp shuffle ThorpFirstReciprocal, ThorpRemaining, PartialPermutation, ThorpRouting, OccupiedOverlap, WeightedSweepMoments, CoordinateTrace, ThorpWeightedCompatibility, SpinAngle, SignedSweepMoment, BinarySweep, CoordinateSweeps, FourRowPermanent	12	● 10 of 12

#	Result (and its Lean challenges)	Papers	Lean
239	Sharp singularity rates for symmetric sign matrices	2	—

Mathematical logic · 6 families, 6 with Lean

#	Result (and its Lean challenges)	Papers	Lean
240	Shelah's eventual categoricity conjecture CHObstruction	2	● 1 of 2 major gap
241	Rigidity of the Turing degrees DegreeRigidity	1	● 1 of 1
242	Single-fold Diophantine representations SingleFold	1	● 1 of 1
243	Choiceless counting does not capture polynomial time ChoicelessPolynomialTime, WitnessedChoice	2	● 2 of 2
244	The partition principle does not imply choice PartitionPrinciple, PartitionConsistency	1	● 1 of 1 minor gap
245	The β-Barendregt–Geuvers–Klop conjecture TypeSystemNormalization	1	● 1 of 1

Group theory · 14 families, 12 with Lean

#	Result (and its Lean challenges)	Papers	Lean
246	Cannon's conjecture ► guide CannonGeometricAction	1	● 1 of 1
247	An infinite finitely presented residually finite 2-group PeriodicGroup	2	● 1 of 2 major gap
248	Thompson's group F is nonamenable ► guide ThompsonNonamenability	1	● 1 of 1
249	A finitely generated counterexample to Eilenberg–Ganea EilenbergGanea	1	● 1 of 1
250	The Boone–Higman conjecture and higher finiteness BooneHigman, SimpleOvergroups, UniversalFInfinity	3	● 3 of 3
251	Amenability, unitarizability, and Ulam stability DixmierAllDiscrete, Dixmier	2	● 1 of 2 minor gap
252	A non-residually-finite torsion-free hyperbolic group TorsionFreeHyperbolic	1	● 1 of 1
253	An infinite finitely presented simple amenable group SimpleAmenable	1	● 1 of 1
254	Classifying spaces and geometric obstructions for Artin groups HarmonicArtin, ArtinCAT0, ArtinParabolicIntersections	3	● 3 of 3
255	Quasi-isometric rigidity of virtually polycyclic groups PolycyclicRecognition	1	● 1 of 1
256	Howie's conjecture on equations over groups Kervaire	2	● 1 of 2 major gap
257	A hyperbolic group with no geometric CAT(0) action HyperbolicObstruction	1	● 1 of 1
258	Gersten's conjecture for one-relator groups	2	—
259	A group without fixed price	1	—

Mathematical physics · 25 families, 17 with Lean

#	Result (and its Lean challenges)	Papers	Lean
260	Spacetime Penrose inequalities and rigidity CKSBondiPenrose	13	● 1 of 13 major gap
261	Localization and delocalization in the Anderson model PlanarAndersonSpectrum	2	● 1 of 2 major gap
262	Sharp one-dimensional Lieb–Thirring inequalities LiebThirring	3	● 1 of 3 minor gap
263	The ionization and generalized ionization conjectures CoulombIonization, CoulombRadii	3	● 2 of 3 major gap
264	Strong cosmic censorship near two-ended Kerr data	3	—
265	The two-dimensional gapped area law	2	—
266	Exactly three mutually unbiased bases in dimension six MUBSix, HadamardCubeFiber	2	● 2 of 2 major gap
267	Positive-temperature Bose–Einstein condensation and quantum depletion HardSphere	5	● 1 of 5 major gap
268	The spin-one Haldane gap	2	—
269	The Laughlin gap and stability under scalar disorder ► guide LaughlinGap, Laughlin, LaughlinFock, LaughlinPlanar	2	● 2 of 2 minor gap
270	Threshold and positive-energy bound states of the BFSS model	2	—
271	Bloch's law and spontaneous ferromagnetic order ► guide Heisenberg	4	● 1 of 4 major gap
272	Entanglement without secret key and the PPT-square conjecture DimensionTenChannel, DimensionTenPair	1	● 1 of 1 major gap
273	The entropy photon-number inequality EntropyPhotonNumber	1	● 1 of 1
274	Moore's parity conjecture for QAC⁰ QACParity, RegularParity, RegularParity, QACParity	2	● 2 of 2
275	QMA-hardness of continuum Coulomb energy ContinuumCoulombHardness, ContinuumCoulombHardness	2	● 2 of 2
276	The classical capacity of generalized amplitude damping AmplitudeDamping	1	● 1 of 1 minor gap
277	Threshold repetition for entangled games EntangledGames	1	● 1 of 1
278	Failure of Kohn–Sham ensemble representation	1	—
279	Exact quantum factoring over a fixed finite gate set ExactQuantumFactoring	1	● 1 of 1
280	Strong locality for strongly rational unitary vertex operator algebras VertexAlgebraNet	1	● 1 of 1 minor gap
281	QAOA optimality for the SK model SKValue, SKFullSupport	2	● 2 of 2 major gap
282	Scale-to-conformal enhancement in four-dimensional QFT	1	—
283	Polynomial-time, constant-error unitary synthesis from a Boolean oracle	1	—
284	The optimal randomized–quantum query exponent	1	—

Operator algebras · 19 families, 14 with Lean

#	Result (and its Lean challenges)	Papers	Lean
285	Counterexamples to Baum–Connes and Kadison–Kaplansky	3	—
286	Arithmetic rigidity of lattice von Neumann algebras	2	—
287	All nonabelian free group factors are isomorphic ▶ guide InterpolatedFactors	1	● 1 of 1
288	Kadison's similarity conjecture ▶ guide KadisonSimilarity, UniformCommutator	1	● 1 of 1
289	The strong Kadison–Kastler conjecture StrongKadisonKastler	3	● 1 of 3
290	Connes' bicentralizer conjecture and relative bicentralizers BoundedRecovery	2	● 1 of 2 major gap
291	Toms–Winter and equivariant Jiang–Su stability UniformGamma	4	● 1 of 4 major gap
292	A counterexample to Kirchberg's norm-ultrapower embedding problem NuclearUltrapower	1	● 1 of 1
293	A counterexample to the hyperinvariant-subspace problem FiniteFactor, BackwardIntertwiners, ContinuousCircleWeight, HyperinvariantSubspaces, IrrationalRotation, ProductBrown, HyperinvariantSubspaces	2	● 2 of 2
294	A counterexample to Kaplansky's quasitrace conjecture KaplanskyQuasitrace, KaplanskyStableFiniteness	1	● 1 of 1
295	The Kadison–Ringrose cohomology conjecture KadisonRingrose	1	● 1 of 1
296	The generator problem for finite factors FactorGeneration, RelativeGeneration	1	● 1 of 1
297	A ZFC counterexample to Naimark's problem Naimark	1	● 1 of 1
298	A counterexample to Voiculescu's free-entropy equality conjecture FiniteEntropySeparation	1	● 1 of 1
299	The Kirchberg–Rørdam character criterion CharacterCriterion	1	● 1 of 1 minor gap
300	The Popa–Vaes quadratic strong-operator paving conjecture	2	—
301	Classification by trace cones after Razak–Jacelon stabilization	1	—
302	The Phillips–Toms formula for minimal integer actions	2	—
303	From ordinary to strong pure infiniteness ExactInfiniteness, IndividualStrongInfiniteness	1	● 1 of 1

Topology · 18 families, 3 with Lean

#	Result (and its Lean challenges)	Papers	Lean
304	The Hilbert–Smith conjecture in every dimension ▶ guide	1	—
305	Counterexamples to disk embedding and Wall's manifold conjecture	3	—
306	The purely cosmetic surgery conjecture for knots in S^3	1	—
307	Failure of rational injectivity for maximal coarse assembly <code>CoarseAssembly</code>	2	● 1 of 2 major gap
308	Smith–Toda complexes at every height with varying primes	2	—
309	The Kervaire invariant problem at the prime three	1	—
310	Quillen's conjecture in rational homology	1	—
311	Chai's invariant-ideal conjecture	1	—
312	The Grothendieck homotopy hypothesis <code>GrothendieckElementaryExpansion</code>	1	● 1 of 1 major gap
313	Finite generation for the $K(n)$-local sphere	1	—
314	The chromatic Smith fixed-point problem for finite p-groups	1	—
315	The four-dimensional Singer conjecture	1	—
316	Curtis's conjecture	1	—
317	Thomason model structures in every strict higher dimension <code>ThomasonModelStructures</code>	1	● 1 of 1
318	Chromatic splitting: counterexamples and filtrations	5	—
319	Counterexamples to the Hahn–Wilson conjecture at height two	1	—
320	A four-dimensional counterexample to Borel rigidity	1	—
321	A counterexample to Wall's finite $D(2)$ conjecture	1	—

Functional analysis · 11 families, 10 with Lean

#	Result (and its Lean challenges)	Papers	Lean
322	Tingley's sphere-isometry problem <code>TingleySphereIsometry</code>	1	● 1 of 1
323	Relative independence of the separable quotient problem <code>SeparableQuotientNegative</code>	1	● 1 of 1 major gap
324	Lipschitz equivalence without linear isomorphism <code>LipschitzEquivalence</code> , <code>C0Absorption</code>	2	● 2 of 2
325	The complete Crouzeix conjecture <code>DirectCrouzeix</code> , <code>CompleteCrouzeix</code> , <code>StructuralCrouzeix</code> , <code>CrouzeixHilbert</code> , <code>HilbertCrouzeix</code>	2	● 2 of 2
326	The cotype–cotype conjecture under the approximation property <code>Cotype</code>	1	● 1 of 1
327	Markov type characterizes superreflexivity <code>MarkovType</code>	1	● 1 of 1
328	Fixed points of nonexpansive maps in reflexive Banach spaces <code>ReflexiveFixedPoints</code>	1	● 1 of 1
329	A counterexample to Pietsch's metric-entropy duality conjecture <code>MetricEntropyDuality</code>	1	● 1 of 1
330	A negative answer to Kalton's Lipschitz-free approximation question <code>Reall1Renorming</code> , <code>DiscreteLipschitzFree</code>	1	● 1 of 1
331	Asymptotic midpoint uniform convexity without asymptotically uniformly convex renorming <code>ForestSpace</code> , <code>MidpointLenses</code> , <code>DiamondDistortion</code> , <code>DaugavetModuli</code> , <code>BoundedTreePotentials</code> , <code>IndependentProducts</code> , <code>RecursivePotentials</code>	7	● 7 of 7
332	Metric Markov cotype two for ℓ_1	1	—

Differential geometry · 29 families, 15 with Lean

#	Result (and its Lean challenges)	Papers	Lean
333	Smooth isometric immersions of surfaces into $\mathbb{R}^4$ SurfaceImmersion	1	● 1 of 1
334	A smooth surface metric with no local immersion in $\mathbb{R}^3$ IsometricImmersion	1	● 1 of 1
335	Gromov's integral scalar-curvature inequality	2	—
336	Spectral scalar curvature and codimension-two width	3	—
337	Cartan–Hadamard isoperimetry and CAT(0) fillings FillingCoefficient, SharpCAT0Filling	2	● 1 of 2 major gap
338	Yau's uniformization conjecture	1	—
339	Katok's entropy rigidity conjecture	1	—
340	A counterexample to the nearby Lagrangian conjecture	1	—
341	Donaldson's hypersymplectic deformation conjecture	1	—
342	Donaldson's tamed-to-compatible conjecture TamingCompatibility	1	● 1 of 1
343	Sharp symplectic ball-packing criteria in higher dimensions BallPacking, BallPackingNecessity	1	● 1 of 1
344	The metric Blaschke conjecture	1	—
345	Infinitely many closed geodesics on spheres and three-manifolds	1	—
346	Sharp singular-set bounds for stationary integral varifolds	2	—
347	Counterexamples to strong forms of Arnold's fixed-point conjecture ArnoldCounterexample	5	● 1 of 5 major gap
348	Nonnegatively curved Einstein four-manifolds and a topological gap EinsteinFour	3	● 1 of 3 major gap
349	The Solomon–Yau least-volume conjecture	1	—
350	Yau's nodal conjecture: surfaces and counterexamples NodalLength, SmoothYau, YauCounterexample	3	● 3 of 3 minor gap
351	Scalar curvature detects four-dimensional Ricci-flow singularities	3	—
352	Finite-time singularity of Calabi flow	1	—
353	The sharp dimension threshold for affine Bernstein rigidity AffineBernstein	2	● 1 of 2 minor gap
354	Isoperimetric regions in the cubic three-torus CubicTorus	1	● 1 of 1
355	Unique tangent flows at the first surface singularity	1	—
356	Gigli's characterization of Alexandrov curvature WeakHessian	2	● 1 of 2 major gap
357	Bi-Lipschitz coordinates at every regular RCD point	1	—
358	A three-manifold with no conjugate points and no nonpositively curved metric ConjugatePoints	1	● 1 of 1
359	Negative Kähler curvature without bounded holomorphic coordinates PinchedKähler	2	● 1 of 2
360	Villani's convexity conjecture and regular optimal transport WeakMTWGlobalSupport, BiholderTransport	2	● 2 of 2
361	Counterexamples to Yau's harmonic dimension bound HarmonicGrowth	2	● 1 of 2 minor gap

Partial differential equations · 16 families, 11 with Lean

#	Result (and its Lean challenges)	Papers	Lean
362	Global smoothness for relativistic Vlasov–Maxwell ► guide VlasovMaxwell	1	● 1 of 1
363	Boltzmann nonuniqueness with exact local conservation BoltzmannNonuniqueness	2	● 1 of 2 major gap
364	Kinetic limits and fluctuations over the Boltzmann lifespan	2	—
365	Joint metric and connection recovery from one boundary patch Conductivity	3	● 1 of 3 major gap
366	The planar Mumford–Shah regularity conjecture	1	—
367	The critical dimension for the one-phase Bernoulli problem BernoulliNonflatInSeven	1	● 1 of 1 major gap
368	The three-dimensional Ball–Evans approximation problem	2	—
369	The hot spots conjecture for simply connected planar domains HotSpots	1	● 1 of 1
370	The subcritical Lane–Emden and Hénon–Lane–Emden conjectures HenonEmden	1	● 1 of 1
371	Stable blowup for a defocusing Schrödinger equation DefocusingNLS	1	● 1 of 1
372	Global uniqueness in smooth isotropic elasticity ElasticityUniqueness	1	● 1 of 1
373	Nonattainment in three-marginal Coulomb transport CoulombCounterexample	1	● 1 of 1 minor gap
374	Sharp one-third stability of optimal transport maps Brenier	1	● 1 of 1
375	De Giorgi's conjecture in dimension eight	1	—
376	Universal computation in forced Navier–Stokes flows SolenoidalSheetPrograms, NavierStokesAlternating, NavierStokesVelocity, BalancedBoxRouting, BalancedThreeStack, ForcedNavierStokesComputation	9	● 4 of 9
377	Interior $C^{1,\alpha}$ regularity for infinity-harmonic functions	1	—

Appendix B Sources

- [GitHub: openai/math](https://github.com/openai/math)
<https://github.com/openai/math>
- [Your fork: oishik1306/math](https://github.com/oishik1306/math)
<https://github.com/oishik1306/math>
- [Unite.AI: OpenAI Releases 722 Math Manuscripts From an Unreleased AI Model](https://www.unite.ai/openai-releases-722-math-manuscripts-from-an-unreleased-ai-model/)
<https://www.unite.ai/openai-releases-722-math-manuscripts-from-an-unreleased-ai-model/>
- [FourWeekMBA: OpenAI Says Internal Model Produced 722 Math Manuscripts](https://fourweekmba.com/ai-openai-says-internal-model-produced-722-math-manuscripts/)
<https://fourweekmba.com/ai-openai-says-internal-model-produced-722-math-manuscripts/>
- [OpenAI Developer Community: First look at mathematics manuscripts from an internal frontier model](https://community.openai.com/t/first-look-at-mathematics-manuscripts-from-an-internal-frontier-model-at-openai/1403886)
<https://community.openai.com/t/first-look-at-mathematics-manuscripts-from-an-internal-frontier-model-at-openai/1403886>
- [arXiv 2608.14673: A Human Audit of OpenAI's AI-Generated Mathematical Proofs](https://arxiv.org/html/2608.14673v3)
<https://arxiv.org/html/2608.14673v3>
- [The Next Web: OpenAI says its next model, Astra, has solved ten open problems](https://thenextweb.com/news/openai-astra-model-ten-math-proofs-non-sofic-groups)
<https://thenextweb.com/news/openai-astra-model-ten-math-proofs-non-sofic-groups>
- [SiliconANGLE: OpenAI's Astra solves 10 long-open math problems](https://siliconangle.com/2026/08/02/openais-astra-solves-10-long-open-math-problems-publishes-proofs/)
<https://siliconangle.com/2026/08/02/openais-astra-solves-10-long-open-math-problems-publishes-proofs/>
- [NPR: AI solved one of math's hardest problems. Humanity learned nothing \(so far\)](https://www.npr.org/2026/09/22/nx-s1-5968588/openai-navier-stokes-problem-mathematicians-learn-little)
<https://www.npr.org/2026/09/22/nx-s1-5968588/openai-navier-stokes-problem-mathematicians-learn-little>
- [The Neuron: Inside OpenAI's Navier–Stokes claim](https://theneuron.ai/news/inside-openais-navierstokes-claim-the-proof-the-ai-effort-and-the-credit-fight/)
<https://theneuron.ai/news/inside-openais-navierstokes-claim-the-proof-the-ai-effort-and-the-credit-fight/>
- [TechCrunch: OpenAI's feud with mathematicians is only escalating](https://techcrunch.com/2026/09/11/openais-feud-with-mathematicians-is-only-escalating/)
<https://techcrunch.com/2026/09/11/openais-feud-with-mathematicians-is-only-escalating/>
- [TechCrunch: OpenAI's embarrassing math \(October 2025\)](https://techcrunch.com/2025/10/19/openais-embarrassing-math/)
<https://techcrunch.com/2025/10/19/openais-embarrassing-math/>
- [Terence Tao: Announcing the Advisory Group on Mathematics and Artificial Intelligence](https://terrytao.wordpress.com/2026/09/21/advisory-group-on-mathematics-and-artificial-intelligence/)
<https://terrytao.wordpress.com/2026/09/21/advisory-group-on-mathematics-and-artificial-intelligence/>
- [Terence Tao: Hexagon](https://terrytao.wordpress.com/2026/10/06/hexagon/)
<https://terrytao.wordpress.com/2026/10/06/hexagon/>

Files in the repository used for this guide: README.md, overview.tex, CONTENTS.md, lean/formalization.yaml, lean/lakefile.lean, lean/ComparatorChallenges/*, lean/OAI/*, and reasoning_traces/irrationality-exponent-of-pi.pdf.